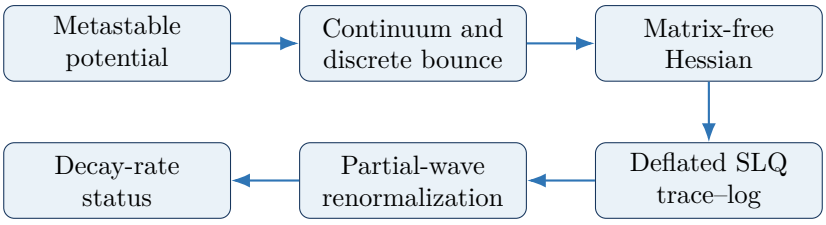
Higgs Fluctuation Solver in 4D
– Beyond the Standard Model –

A Technical Guide to the Unified Bounce, Hessian,
Stochastic-Lanczos, and Continuum-Determinant Pipeline



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In Vero Vacuo

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Source program: `Higgs_fluctuation_solver_stabilized.py`
Production calculation: $L = 26$, $h = 0.8$, global-$O(4)$ scalar model
In Vero Vacuo project website

Preface and Scope

This document explains the architecture, physics, numerical algorithms, and outputs of the unified source file

`Higgs_fluctuation_solver_stabilized.py`

The program is a unified research solver for a four-component scalar field in four-dimensional Euclidean space, with quadratic $SU(2)$ gauge diagnostics and a nonlinear lattice-vertex generation mode. It contains ten execution modes: analytic validation, $SU(2)$ prerequisite validation, a regulated gauge–Goldstone–ghost determinant scan, an $SU(2)$ counterterm stage, nonlinear SU(2) vertex-table generation, exploratory research spectra, discrete bounce relaxation, stochastic Lanczos quadrature (SLQ), scalar continuum partial-wave renormalization, and a full scalar pipeline. This revision documents the source and production-recovery work through October 2, the single-core validation audits completed on October 5, the matching procedure updates through October 7, 2026, the October 7 implementation and independent audit of nonlinear SU(2) lattice vertices, and the October 8 fixed full-gauged routing/projector deck and independent audit. It also records a first gauge-linked radial four-point insertion pilot with a regulated continuum–lattice comparison. The full multi-point integrations, complete Ward/Slavnov–Taylor system, regulator-cancellation study, and finite full-gauged coefficients remain outstanding.

The broader motivation for this research program, together with its companion project on quantum computing, is described on the In Vero Vacuo project website. The site provides project-level context complementary to the technical derivations and numerical results documented here.

The title places the work in a "Beyond the Standard Model" research context, because metastable scalar potentials, additional scalar operators, and vacuum decay are common ingredients in extensions of the Standard Model. The implemented production calculation itself is more specific: it is a *global* $O(4)$ scalar model. The updated source additionally contains the quadratic gauge–Goldstone and ghost operators around a zero-gauge-field radial background, including a link-derived Hessian in positive background R_ξ gauges. The source now also evaluates regulated finite-lattice gauge–Goldstone–ghost determinant ratios, treats the false-vacuum constant modes and their Gaussian measure normalization explicitly, and constructs the local counterterm basis with derived continuum pole residues. Scalar-only radial one-loop finite coefficients have now been calculated at the stated benchmark. The full gauged finite matching constants and controlled continuum and volume limits remain incomplete, so a physical gauged prefactor has not yet been calculated.

Chapter 9 incorporates the October 5 $L = 6$ pilot, increased-probe convergence and complete-multiplet studies, and the fresh $L = 3, 4$ nonconstant-background prerequisite audit. It distinguishes recorded follow-up results from the fresh audit and identifies which raw artifacts remain available. These tests do not determine finite matching coefficients or a physical gauged decay rate.

The October 6 matching review is recorded in `SU2_Matching_Coefficient_Progress_Report_20261006.tex` and summarized in Section 9.7. It confirms that the radial basis, pole residues, and coefficient interface are ready. The October 7 matching update reports finite scalar-only radial one-loop coefficients. As of the October 8 update, the full-gauged routing/projector setup is fixed and audited; a first gauge-linked four-point sector pilot is evaluated, while the complete multi-point integrations and finite coefficients remain outstanding. The October 8 update to `SU2_Next_Steps_Finite_Matching_20261006.tex`

adds a proposed nonlinear Wilson-action completion and one-loop projection formulas. The scalar-link, Wilson-gauge, and ghost vertex generators for that action have since been implemented in `su2_lattice_vertices.py`, connected to the solver's `su2_vertices` mode, and independently checked against finite transformations and separate action expansions. The update also records one scalar quartic tadpole as a normalization check. A separate October 8 evaluator integrates the gauge-linked contribution to one fixed-route four-radial-Higgs projection and performs a four-point IR scan. Neither this pilot nor the vertex implementation completes full gauged matching.

The foundational progress report, including the adopted EFT matching prescription, is incorporated as Appendix D. Its body is shared with the standalone report through `SU2_Physical_Prefactor_Progress_Content.tex`, so the two presentations use the same report text, equations, tables, and validation graph.

The numerical figures reproduced here were generated by the solver and its companion reports. The conceptual sequence also incorporates material from the local slide sources `bounce_solution.tex`, `one_loop_fluctuations.tex`, `functional_determinants.tex`, and `Lattice_decay_research.tex`.

Scope and interpretation

The finite result quoted in this guide is the continuum-renormalized fluctuation determinant of the global scalar theory. It is not a completed electroweak or gauged-$SU(2)$ vacuum-decay prediction. Three internal orientation collective-coordinate normalizations also remain to be supplied if the symmetry is treated as global.

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Chapter 1

The Physical and Computational Problem

1.1 Euclidean false-vacuum decay

For a real four-component scalar field $\mathbf{\Phi}(x)$, define its radial amplitude $\rho = (\mathbf{\Phi}\cdot\mathbf{\Phi})^{1/2}$. The Euclidean action is

$$S_E[\mathbf{\Phi}] = \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\mathbf{\Phi}\cdot\partial_\mu\mathbf{\Phi} + V(\rho)\right]. \tag{1.1}$$

The semiclassical decay rate per unit four-volume has the schematic form

$$\frac{\Gamma}{V_4} = \left(\frac{B}{2\pi}\right)^2 \left|\frac{\det{}' H_B}{\det H_F}\right|^{-1/2} e^{-B} \times \mathcal{J}_{\text{internal}} \times [1 + \text{higher loops}], \tag{1.2}$$

where $B = S_E[\rho_B] - S_E[0]$, H_B is the Hessian around the bounce, H_F is the false-vacuum Hessian, and the prime means that collective coordinates are omitted from the determinant [1, 2].

1.2 Metastable scalar potential

The code uses the polynomial

$$V(\rho) = -\frac{\mu^2}{2}\rho^2 + \frac{\lambda}{4}\rho^4 + \frac{\gamma}{6}\rho^6 + \frac{\delta}{8}\rho^8. \tag{1.3}$$

The production parameter set is

$$\mu^2 = -0.40, \qquad \lambda = -2.00, \qquad \gamma = 0.80, \qquad \delta = 0.08. \tag{1.4}$$

With the sign convention in Eq. (1.3), the curvature at the false vacuum is $m_F^2 = -\mu^2 = 0.4 > 0$. The negative quartic creates a barrier, while the positive sixth- and eighth-order terms stabilize the potential at large field amplitude.

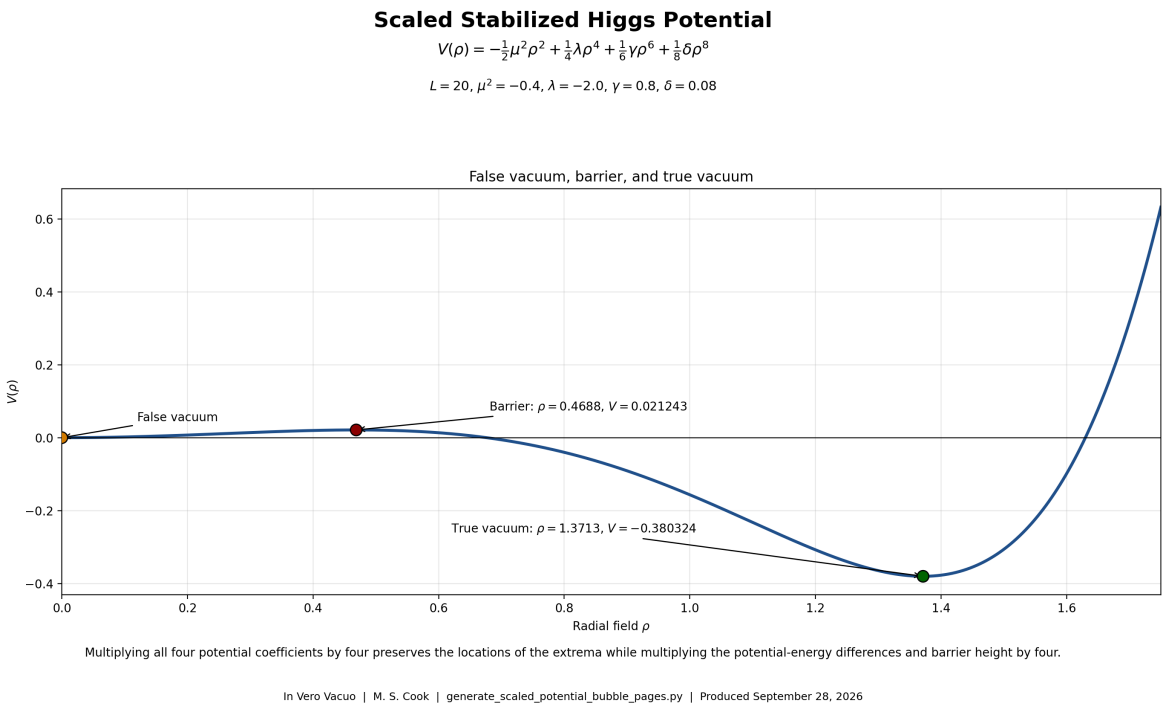


Figure 1.1: Scaled metastable potential used for the production calculation. The false vacuum is at $\rho = 0$, separated by a barrier from the deeper nonzero minimum.

Implementation note

Nonzero stationary points are found by solving a cubic in $s = \rho^2$, $\delta s^3 + \gamma s^2 + \lambda s - \mu^2 = 0$. The radial curvature identifies barrier maxima and true-vacuum minima before any bounce shooting is attempted.

```
    for _stage, _overrides in _stage_environments:
        _environment = _pipeline_os.environ.copy()
        _environment.update(_overrides)
        _environment['HIGGS_RUN_MODE'] = _stage
        print(f'\n=== Running unified pipeline stage: {_stage} ===', flush=True)
        _pipeline_subprocess.run(
            [_pipeline_sys.executable, __file__],
            check=True,
            env=_environment,
        )
    raise SystemExit(0)

if _PIPELINE_MODE == 'relax':
    """Relax a sampled continuum bounce to a discrete lattice stationary point.

    The nonlinear equation solved is

        -Delta_h rho(x) + V'(rho(x)) = 0

    on a periodic four-dimensional lattice.  Newton corrections are obtained with
    matrix-free MINRES using the longitudinal Hessian.  A residual-norm line search
    keeps the iteration in the basin of the nontrivial saddle rather than blindly
    taking a full Newton step.
    """

```

```
    import json
    import os
    import sys
    import time
    from datetime import datetime
    from pathlib import Path

    import matplotlib.pyplot as plt
    import numpy as np
    import scipy.sparse.linalg as spla
    from matplotlib.backends.backend_pdf import PdfPages
    from scipy.integrate import solve_ivp


    L = int(os.getenv('HIGGS_RELAX_L', '26'))
    SPACING = float(os.getenv('HIGGS_RELAX_SPACING', '0.8'))
    MAX_NEWTON = int(os.getenv('HIGGS_RELAX_MAX_NEWTON', '14'))
    SKIP_SPECTRUM = os.getenv('HIGGS_RELAX_SKIP_SPECTRUM', '0') == '1'
    OUTPUT_DIRECTORY = Path(os.getenv(
        'HIGGS_RELAX_OUTPUT', f'discrete_bounce_relaxation_L{L}_h{SPACING:g}'
    ))
    OUTPUT_DIRECTORY.mkdir(exist_ok=True)
    CHECKPOINT = OUTPUT_DIRECTORY / 'relaxed_bounce_checkpoint.npz'

    MU_SQ = -0.40
    LAMBDA = -2.00
    GAMMA = 0.80
    DELTA = 0.08
    N = L**4
    SHAPE = (L,) * 4


    def potential(rho):
```

Listing 1.1: Potential, derivatives, and stationary-point classification in the unified source.

Chapter 2

Program Architecture and Execution Modes

2.1 Unified dispatcher

The merged program begins with a mode dispatcher. Most specialized numerical implementations are embedded directly in the source file; the nonlinear vertex rules are maintained in the imported module `su2_lattice_vertices.py`. The `full` mode launches the relaxation, SLQ, and continuum stages sequentially with a consistent production parameter set. It is a scalar pipeline: setting `HIGGS_SU2_GAUGE=1` in `full`, `relax`, `slq`, or `continuum` now raises an explicit error instead of silently producing scalar-only results. The independent `su2_validation`, `su2_determinant`, `su2_counterterms`, and `su2_vertices` branches run outside the scalar pipeline. This separation prevents a regulated or unmatched gauge result from being silently reported as part of the physical scalar rate.

```
        ),
        "probe_csv": str(probe_file), "summary_csv": str(aggregate_file),
        "low_modes_csv": str(low_file)
    }
    with open(output / "determinant_summary.json", "w", encoding="utf-8") as stream:
        json.dump(summary, stream, indent=2)

    with PdfPages(output / "SU2_Determinant_Gauge_Scan.pdf") as report:
        figure, axes = plt.subplots(1, 2, figsize=(12, 7))
        for shift in shifts:
            selected = [row for row in aggregates if row["steps"] == final_steps
                        and row["ir_shift"] == shift
                        and row["zero_mode_policy"] == "primed_normalized"]
            axes[0].errorbar(
                [row["xi"] for row in selected],
                [row["mean_log_ratio"] for row in selected],
                yerr=[row["standard_error"] for row in selected], marker="o",
                label=f"shift={shift:g}"
            )
        axes[0].set(xlabel=r"Gauge parameter $\xi$", ylabel="Three-generator log ratio"
            ,
                    title="Constant modes removed; measure normalized")
        axes[0].grid(True, alpha=0.3)
        axes[0].legend()
        for gauge_xi in gauge_parameters:
            selected = [row for row in aggregates if row["steps"] == final_steps
                        and row["xi"] == gauge_xi
                        and row["zero_mode_policy"] == "primed_normalized"]
            axes[1].errorbar(
```

```
                [row["ir_shift"] for row in selected],
                [row["mean_log_ratio"] for row in selected],
                yerr=[row["standard_error"] for row in selected], marker="o",
                label=rf"$\xi={gauge_xi:g}$"
            )
        axes[1].set_xscale("log")
        axes[1].set(xlabel="IR mass-squared shift", ylabel="Three-generator log ratio",
```

Listing 2.1: Top-level execution-mode dispatcher.

Table 2.1: Execution modes in the unified solver.

Mode	Purpose	Principal output
`validation`	Exactly solvable periodic-lattice test of matrix-free operators and low spectra.	PASS/FAIL diagnostics
`su2_validation`	Link-Hessian Ward identities, zero-coupling limit, homogeneous Fourier spectra, and optional stationary-bounce checks.	JSON and validation PDF
`su2_determinant`	Regulated coupled gauge–Goldstone and ghost determinant scan over positive R_ξ gauges, with exact and deflated-SLQ backends.	CSV, JSON, low modes, and gauge-scan PDF
`su2_counterterms`	Evaluate the local radial EFT basis, derive the continuum pole residues, and apply supplied finite matching coefficients.	Basis CSV, JSON summary, and matching PDF
`su2_vertices`	Generate nonlinear scalar-link, Wilson-gauge, and ghost momentum/color tables for configured momentum routings.	Compressed NPZ tables and JSON metadata
`research`	Sample a stationary or Gaussian profile and calculate a low-mode Hessian spectrum; optionally add $SU(2)$ diagnostics.	CSV tables and two/three-page PDF
`relax`	Solve the nonlinear discrete lattice equation with Newton–MINRES.	Relaxed field checkpoint
`slq`	Estimate the full bare lattice trace–log with deflated stochastic Lanczos quadrature.	Checkpointed probe statistics
`continuum`	Evaluate the renormalized primed determinant with radial partial waves and Gel'fand–Yaglom ratios.	Finite determinant summary
`full`	Run relaxation, SLQ, and continuum renormalization in order.	Complete scalar calculation set

2.2 Nonlinear lattice SU(2) vertex generation

The nonlinear action remains a proposed matching convention rather than the action used by the scalar production pipeline. Its perturbative vertices are now available as reusable functions in `su2_lattice_vertices.py` and through `HIGGS_RUN_MODE=su2_vertices`. Its principal API is:

- `scalar_link_vertex` and `wilson_gauge_vertex`;

- `ghost_gauge_vertex` and `ghost_scalar_vertices`;
- `faddeev_popov_matrix` and `generate_one_loop_vertex_tables`.

The implementation keeps the specified logarithmic link coordinate $U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a]$, midpoint Fourier phases, and all-incoming momentum convention. It provides scalar-link tensors with two Higgs legs and $r = 0, 1, 2$ gauge legs; Wilson-plaquette tensors with $n = 2, 3, 4$ gauge legs; gauge–ghost tensors with $r = 0, 1, 2$ gauge legs; local scalar-dependent ghost tensors; and the nonzero-background sparse Faddeev–Popov matrix. The scalar potential derivative rules remain as documented in the matching report.

The table-generation mode writes compressed arrays and a JSON sidecar. The default outputs are `su2_nonlinear_vertex_tables_20261007_conserving.npz` and `su2_nonlinear_vertex_tables_20261007_conserving.json`; existing files are not overwritten. These tables include every color and polarization component for their recorded sample momenta, not a scan over every possible external momentum. Reusable vertex functions accept other momentum routings. For example:

```
HIGGS_RUN_MODE=su2_vertices python Higgs_fluctuation_solver_stabilized.py
```

Listing 2.2: Generate the configured nonlinear SU(2) vertex tables.

The independent audit is run separately with `su2_nonlinear_vertex_audit.py`. It compares scalar-link coefficients with finite differences of the exact link exponentials, checks Wilson coefficients against an independent ordered-word expansion and the quadratic transverse kernel, checks ghost vertices through finite transformations in the logarithmic link coordinate, and compares the nonzero-background FP matrix with finite gauge transformations. At $a = 0.8$, $g_0 = 0.65$, and $\xi = 1$, the maximum discrepancies were 6.204×10^{-9} for the complete scalar-link table, 7.216×10^{-16} for the Wilson quadratic kernel, zero for the complete Wilson tensor comparison at the tested routings, 2.398×10^{-6} for the complete gauge–ghost table, and 3.234×10^{-10} absolute (1.679×10^{-11} relative) for the nonzero-background FP matrix. The finite-transformation audit also checks local scalar-ghost tensors and gauge-invariance identities. These are implementation tolerances, not matching uncertainties.

The vertex audit validates rule construction, not the complete gauged matching calculation. Fixed nonexceptional projection kinematics are now available, and a separate evaluator has integrated one gauge-linked four-Higgs sector with a common IR mass regulator. The generators are still not connected to a complete multi-point matching integrator or physical determinant calculation. The remaining integrated sectors, regulator cancellation, full Ward/Slavnov–Taylor checks, and gauge/parameter matching are outstanding. Accordingly, the coefficient template and full-matching status must remain unset/false.

2.3 Data flow

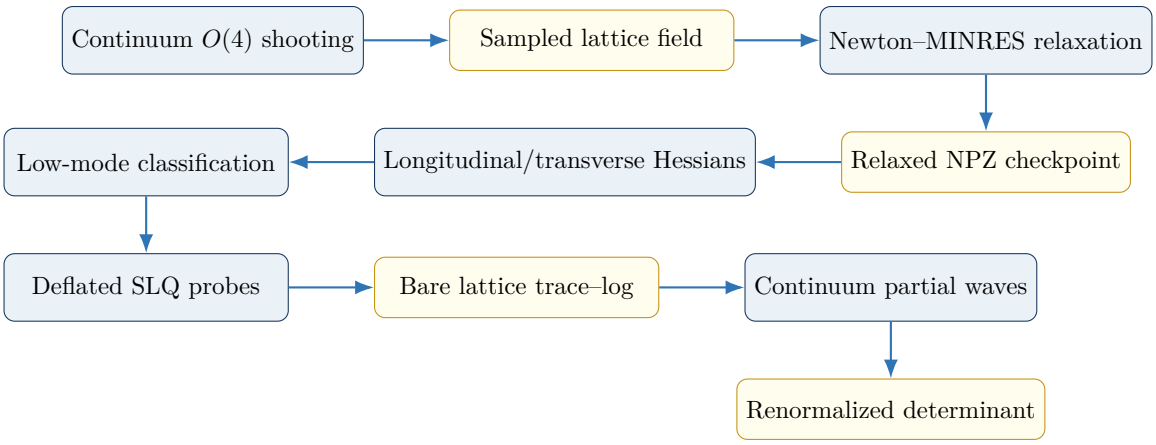


Figure 2.1: Computational data flow. The bare lattice and continuum calculations are complementary: SLQ diagnoses the complete finite-lattice trace, while the partial-wave calculation supplies the regulator-subtracted finite result.

Chapter 3

Lattice Geometry, Units, and Matrix-Free Operators

3.1 Four-dimensional lattice

For a periodic hypercubic lattice with L sites per direction,

$$N_{\text{sites}} = L^4, \qquad N_{\text{dof}} = 4L^4. \tag{3.1}$$

At $L = 26$, each scalar channel has 456,976 sites and the full four-component fluctuation space has 1,827,904 degrees of freedom. An explicit dense Hessian is therefore impossible, and even an assembled sparse matrix is unnecessary.

The periodic nearest-neighbor negative Laplacian is

$$(-\Delta_h f)_x = \frac{1}{h^2}\left[8f_x - \sum_{\mu=1}^{4}\left(f_{x+\hat{\mu}} + f_{x-\hat{\mu}}\right)\right]. \tag{3.2}$$

It is applied by reshaping a vector into an L^4 array and using `numpy.roll`. This reduces permanent operator storage to local potential diagonals and working vectors.

```
        diagonal = (
            longitudinal_curvature(field)
            if channel == 'longitudinal' else transverse_curvature(field)
        )

        def matvec(vector):
            grid = np.asarray(vector).reshape(SHAPE)
            return (negative_laplacian(grid) + diagonal * grid).ravel()

        return spla.LinearOperator(
            shape=(N, N), matvec=matvec, rmatvec=matvec, dtype=np.float64
        )


    def continuum_initial_guess():
        roots = np.roots([DELTA, GAMMA, LAMBDA, -MU_SQ])
        points = []
        for root in roots:
            if abs(root.imag) < 1.0e-10 and root.real > 0.0:
                rho_sq = float(root.real)
                rho = np.sqrt(rho_sq)
                curvature = 2.0 * rho_sq * (
```

```
                    LAMBDA + 2.0 * GAMMA * rho_sq + 3.0 * DELTA * rho_sq**2
                )
                points.append((rho, potential(rho), curvature))
        rho_barrier = min(point[0] for point in points if point[2] < 0.0)
        rho_true = min(
            (point for point in points if point[2] > 0.0),
            key=lambda point: point[1],
        )[0]

        def rhs(radius, state):
            rho, rho_prime = state
            return [rho_prime, potential_derivative(rho) - 3.0 * rho_prime / radius]

        def overshoots(rho_center):
            epsilon = 1.0e-6
            force = potential_derivative(rho_center)

            def crossing(radius, state):
                return state[0]

            crossing.terminal = True
            crossing.direction = -1
            solution = solve_ivp(
                rhs, (epsilon, 80.0),
                [rho_center + force * epsilon**2 / 8.0,
                 force * epsilon / 4.0],
                events=crossing, rtol=3.0e-10, atol=1.0e-12, max_step=0.03,
            )
            return bool(solution.t_events[0].size)

        lower = rho_barrier * (1.0 + 1.0e-8)
        upper = rho_true * (1.0 - 1.0e-12)
```

Listing 3.1: Matrix-free periodic Laplacians and scalar Hessian construction.

3.2 Lattice-to-physical scaling

The exploratory research mode includes an explicit lattice spacing a and uses

$$\Phi_{\mathrm{phys}} = \frac{\Phi_{\mathrm{lat}}}{a}, \quad m^2_{\mathrm{phys}} = \frac{m^2_{\mathrm{lat}}}{a^2}, \quad V_{4,\mathrm{phys}} = (La)^4, \quad \Lambda_{\mathrm{UV}} \sim \frac{\pi}{a}. \tag{3.3}$$

The production relaxation and SLQ stages instead work in dimensionless lattice units with a refinement spacing $h = 0.8$. The continuum matching stage uses m_F as the reference scale. These two spacing conventions must not be silently identified. The production gauged benchmark is stated in scaled units; an explicit physical energy conversion is still needed for a rate in GeV^4.

Chapter 4

Analytic Validation Before Production

4.1 Why validation is separate

The default mode replaces the bounce by a homogeneous periodic problem whose Fourier spectrum is known exactly. This tests indexing, periodic boundaries, component ordering, operator symmetry, eigensolver sorting, and degenerate multiplet counting without relying on the difficult bounce calculation.

For momentum components $p_\mu = 2\pi n_\mu / L$, the lattice kinetic eigenvalue is

$$\widehat{p}^2 = \frac{1}{a^2} \sum_{\mu=1}^{4} (2 - 2\cos p_\mu) \,. \tag{4.1}$$

The homogeneous Hessian eigenvalues are $\widehat{p}^2 + U_L$ and $\widehat{p}^2 + U_T$, with one longitudinal and three transverse copies.

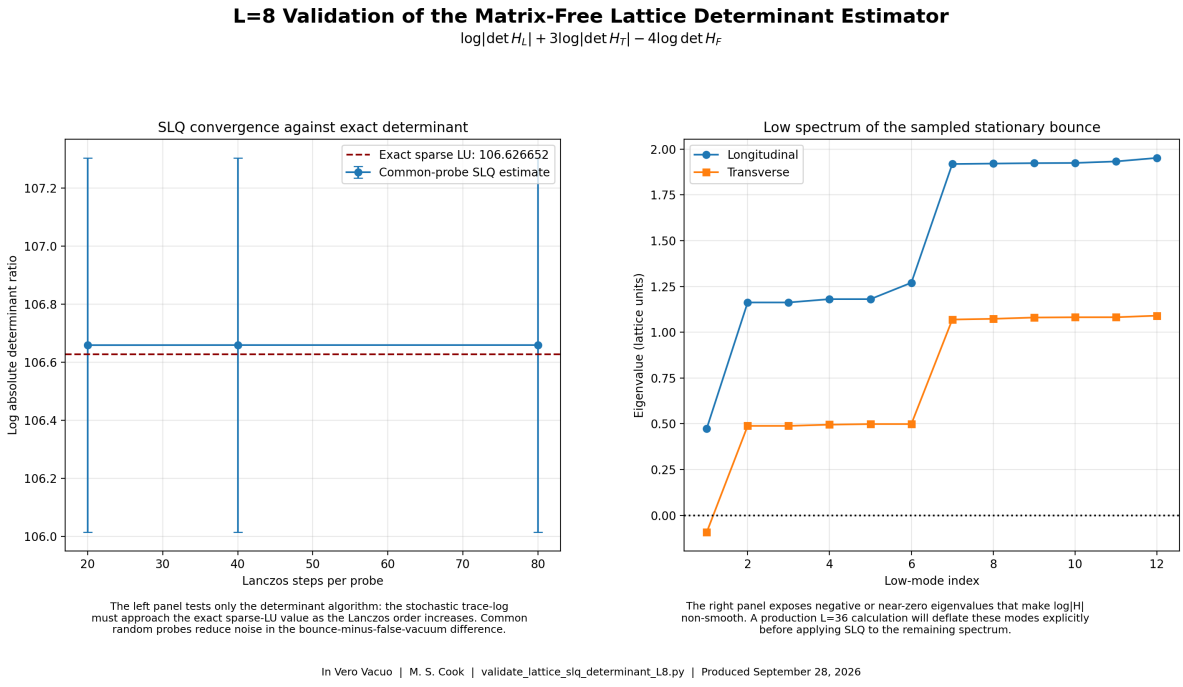


Figure 4.1: Small-lattice validation of the stochastic-Lanczos determinant machinery against a direct reference. Production calculations are accepted only after the matrix-free construction passes analytic checks.

Table 4.1: Core validation diagnostics used by the program.

Diagnostic	Failure detected
Matrix-free/reference product error	Incorrect stencil or component layout
Symmetry residual	Inconsistent `matvec`/`rmatvec` or indexing
Laplacian row-sum residual	Broken periodic constant mode
Low-spectrum error	Eigensolver or operator error
Multiplet multiplicities	Lost degeneracies or incomplete spectral window

4.2 New SU(2) prerequisite validation

The `su2_validation` mode tests the quadratic link-action operator defined in Chapter 7. On a 3^4 grid it checks the exact off-shell lattice Ward identity at $\xi = 0.5, 1, 2$, the $g = 0$ decoupling limit, self-adjointness, and complete homogeneous Fourier spectra. Its optional `HIGGS_SU2_BOUNCE_FILE` input repeats the identity tests on the production $L = 26$, $h = 0.8$ field using the scaled potential.

Table 4.2: Measured SU(2) prerequisite checks, September 30, 2026.

Check	Result
Small-grid off-shell Ward identity	Relative errors below 3.3×10^{-16}
Stationary-bounce off-shell identity	Relative errors below 2.8×10^{-16}
Stationary-bounce on-shell identity	Relative errors below 5.6×10^{-14}
Homogeneous Fourier spectra	Relative errors below 9.3×10^{-16}
Zero-coupling decoupling	Zero measured relative error
Homogeneous normalized one-loop expression	Gauge-parameter spread 5.12×10^{-13}

The homogeneous determinant test includes the gauge-fixing normalization and uses a positive constant broken background. It is not a bounce/false-vacuum determinant ratio. These prerequisites therefore do not establish gauge independence of a completed decay prefactor.

Chapter 5

The Continuum $O(4)$ Bounce

5.1 Radial equation and shooting method

The least-action bounce may be chosen $O(4)$ symmetric under the standard assumptions for scalar false-vacuum decay [1, 3]. It depends only on $r = (x_1^2 + x_2^2 + x_3^2 + x_4^2)^{1/2}$ and satisfies

$$\rho''(r) + \frac{3}{r}\rho'(r) = V'(\rho), \qquad \rho'(0) = 0, \qquad \rho(\infty) = 0. \tag{5.1}$$

The $3/r$ term is Euclidean radial friction. The code integrates from a small $\epsilon > 0$ using the regular series

$$\rho(\epsilon) = \rho_0 + \frac{V'(\rho_0)}{8}\epsilon^2, \qquad \rho'(\epsilon) = \frac{V'(\rho_0)}{4}\epsilon. \tag{5.2}$$

A bisection search adjusts ρ_0 between undershooting and overshooting solutions until the separatrix approaching the false vacuum is isolated.

```
    )
    u_transverse = (
        -MU_SQ + LAMBDA * rho_sq
        + GAMMA * rho_sq**2 + DELTA * rho_sq**3
    )
    u_false = np.full(SHAPE, FALSE_MASS_SQ)


    def negative_laplacian(field):
        result = 8.0 * field
        for axis in range(4):
            result -= np.roll(field, 1, axis=axis)
            result -= np.roll(field, -1, axis=axis)
        return result / SPACING**2


    def make_operator(diagonal):
        diagonal = np.asarray(diagonal, dtype=float)

        def matvec(vector):
            field = np.asarray(vector).reshape(SHAPE)
            return (negative_laplacian(field) + diagonal * field).ravel()

        return spla.LinearOperator(
            shape=(N, N), matvec=matvec, rmatvec=matvec, dtype=np.float64
        )


```

```
    h_longitudinal = make_operator(u_longitudinal)
    h_transverse = make_operator(u_transverse)
    h_false = make_operator(u_false)


    def load_low_modes(operator, channel):
        cache = OUTPUT_DIRECTORY / f'low_modes_{channel}_k{DEFLATED_MODES}.npz'
        if cache.exists():
            cached = np.load(cache)
            print(f'Loaded {channel} low-mode cache.', flush=True)
            return cached['eigenvalues'], cached['eigenvectors']
        print(f'Computing {DEFLATED_MODES} lowest {channel} modes...', flush=True)
        values, vectors = spla.eigsh(
            operator, k=DEFLATED_MODES, which='SA',
            ncv=max(80, 2 * DEFLATED_MODES + 1), tol=2.0e-9,
            maxiter=30000,
        )
        order = np.argsort(values)
        values = values[order]
        vectors = vectors[:, order]
        np.savez_compressed(cache, eigenvalues=values, eigenvectors=vectors)
        return values, vectors


    long_values, long_vectors = load_low_modes(h_longitudinal, 'longitudinal')
    trans_values, trans_vectors = load_low_modes(h_transverse, 'transverse')

    # Exact mode classification established by the relaxed Hessian calculation:
    # longitudinal index 0 is the physical negative mode; indices 1:5 are the
    # four lattice-lifted translation collective coordinates.  The transverse
    # mode of smallest absolute eigenvalue is one orientation direction, repeated
    # over the three internal transverse components.
    translation_indices = np.arange(1, 1 + TRANSLATION_MODES)
    orientation_index = int(np.argmin(np.abs(trans_values)))
    long_keep = np.ones(DEFLATED_MODES, dtype=bool)
    long_keep[translation_indices] = False
    trans_keep = np.ones(DEFLATED_MODES, dtype=bool)
    trans_keep[orientation_index] = False
    exact_low_long = float(np.sum(np.log(np.abs(long_values[long_keep]))))
    exact_low_trans = float(np.sum(np.log(np.abs(trans_values[trans_keep]))))
    zero_mode_normalization = (
        TRANSLATION_MODES + ORIENTATION_MODES
    ) * np.log(FALSE_MASS_SQ)


    def lanczos_tridiagonal(operator, vector, deflation_vectors=None):
        if deflation_vectors is not None:
            vector = vector - deflation_vectors @ (deflation_vectors.T @ vector)
        norm = float(np.linalg.norm(vector))
        previous = np.zeros_like(vector)
        current = vector / norm
        beta_previous = 0.0
        diagonal = []
```

Listing 5.1: Continuum bounce equation, shooting bracket, and physical-tail event.

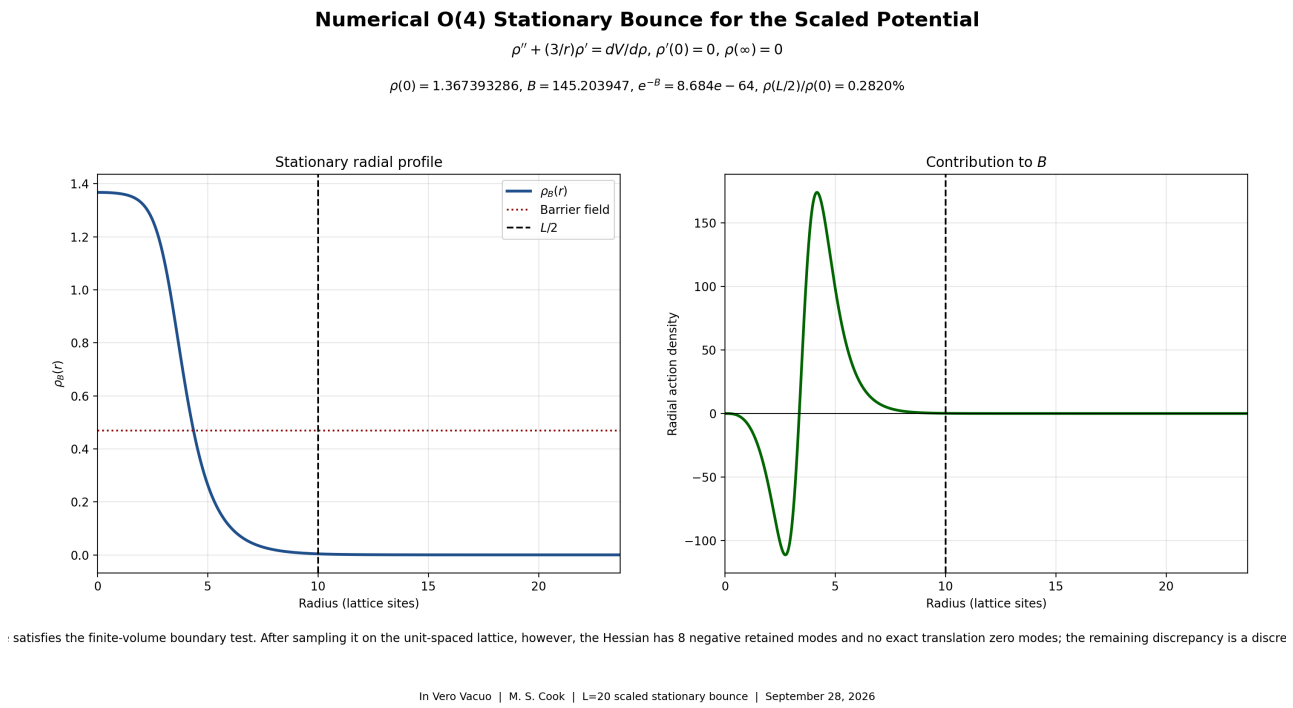


Figure 5.1: Stationary radial bounce and its sampling on the finite periodic lattice. The boundary residual is approximately 0.282%, below the adopted one-percent finite-volume criterion.

5.2 Bounce action

For the radial solution,

$$B = 2\pi^2 \int_0^\infty r^3 \left[\frac{1}{2}\rho_B'(r)^2 + V(\rho_B(r)) - V(0)\right] \mathrm{d}r. \tag{5.3}$$

The production calculation gives

$$B = 145.2039474492, \qquad e^{-B} = 8.6841 \times 10^{-64}. \tag{5.4}$$

The integration terminates when the field reaches its physical false-vacuum tail; extending an inaccurate dense-output interpolant beyond this point would contaminate the action.

Chapter 6

Discrete-Action Bounce Relaxation

6.1 Why continuum sampling is insufficient

A continuum stationary solution sampled on a coarse hypercubic lattice is not automatically stationary under the discrete action. The mismatch lifts the translation modes and can introduce artificial negative deformation modes. The production pipeline therefore solves

$$F[\rho] \equiv -\Delta_h \rho + V'(\rho) = 0 \tag{6.1}$$

directly on the periodic grid.

6.2 Newton–MINRES iteration

At iteration k, the Newton correction obeys

$$H_L[\rho_k], \delta\rho_k = -F[\rho_k], \qquad H_L = -\Delta_h + V''(\rho_k). \tag{6.2}$$

The linear system is solved by matrix-free MINRES because the longitudinal Hessian is symmetric but indefinite near the saddle. A backtracking line search selects α so that

$$\|F[\rho_k + \alpha\delta\rho_k]\|_2 < \|F[\rho_k]\|_2. \tag{6.3}$$

Every accepted step is checkpointed in compressed NPZ form.

```
            xi_label = str(gauge_xi).replace(".", "p")
            count = min(modal_modes, coupled_values.size)
            metadata_path, _, _ = export_su2_modal_dataset(
                modal_output, f"homogeneous_L3_xi_{xi_label}_coupled_A_chi",
                coupled, coupled_values[:count], coupled_vectors[:, :count],
                sector="coupled_A_chi",
                background={
                    "type": "homogeneous_true_vacuum", "field_array": homogeneous_rho,
                    "rho": float(np.sqrt(true_rho_sq)), "goldstone_curvature": 0.0
                },
                lattice={"shape": list(shape), "spacing": spacing,
                         "boundary_conditions": "periodic"},
                parameters={"g": gauge_coupling, "xi": gauge_xi,
                            "ir_shift": 0.0, "generator_multiplicity": 3},
                explicit_operator=(coupled_matrix if coupled.shape[0]
                                   <= modal_max_operator_dim else None)
            )
            modal_datasets.append(str(metadata_path))
```

Listing 6.1: Discrete residual and channel Hessian used by Newton–MINRES.

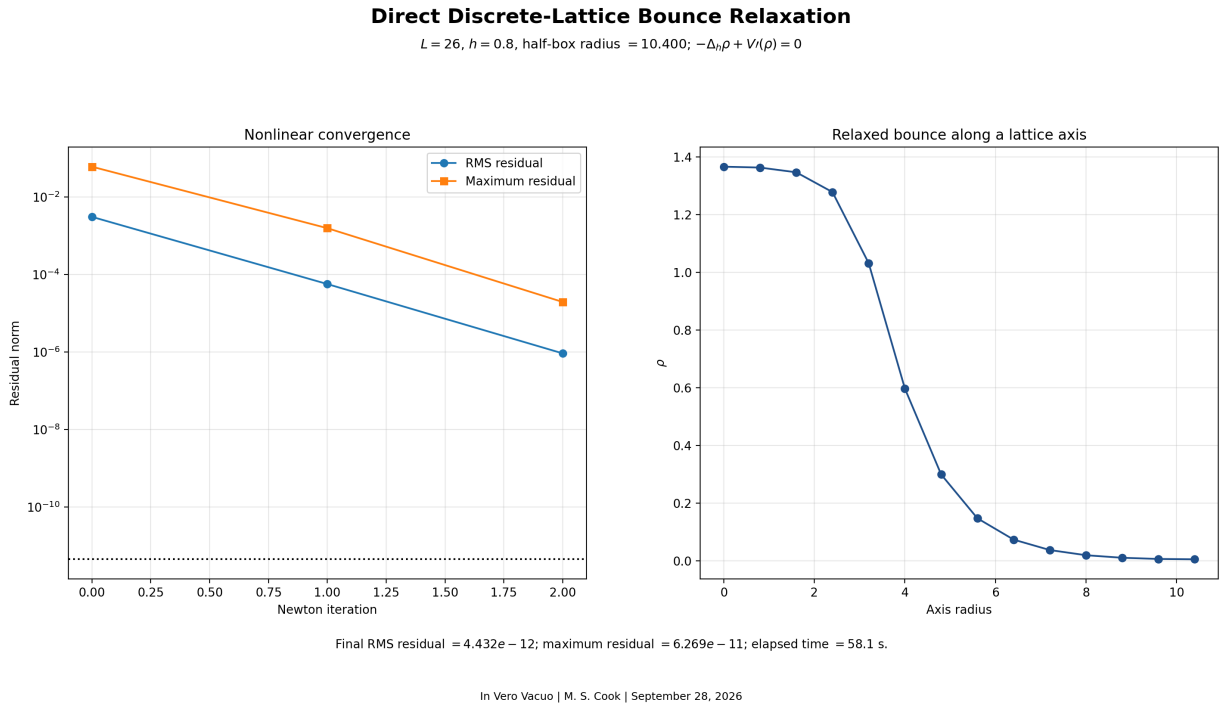


Figure 6.1: Nonlinear convergence and the final relaxed bounce along a lattice axis. The final RMS equation-of-motion residual is 4.43×10^{-12}.

Key result

For $L = 26$ and $h = 0.8$, the relaxed field has RMS residual 4.43266×10^{-12} and maximum residual 6.30906×10^{-11} in the regenerated September 30 field. The identity $H_T \rho = F[\rho]$ is satisfied to the same accuracy, independently confirming the internal-orientation zero mode.

Chapter 7

Hessian Construction and Fluctuation Channels

7.1 Longitudinal and transverse curvatures

For an $O(4)$-invariant potential, fluctuations separate into one radial longitudinal channel and three degenerate transverse channels. On the bounce,

$$U_L(\rho) = -\mu^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \tag{7.1}$$

$$U_T(\rho) = -\mu^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6. \tag{7.2}$$

The scalar channel operators are

$$H_L = -\Delta_h + U_L(x), \qquad H_T = -\Delta_h + U_T(x), \qquad H_F = -\Delta_h + m_F^2. \tag{7.3}$$

Only one transverse operator must be diagonalized; its determinant contribution is multiplied by three.

7.2 Spectral meaning of the special modes

The physical bounce spectrum and its treatment by collective coordinates follow the standard semiclassical construction [2]. It should contain:

1. one negative radial mode, producing the imaginary part associated with decay;
2. four translation collective modes, one for each Euclidean coordinate;
3. three internal-orientation collective modes for global $O(4) \to O(3)$;
4. a positive continuum above these special modes.

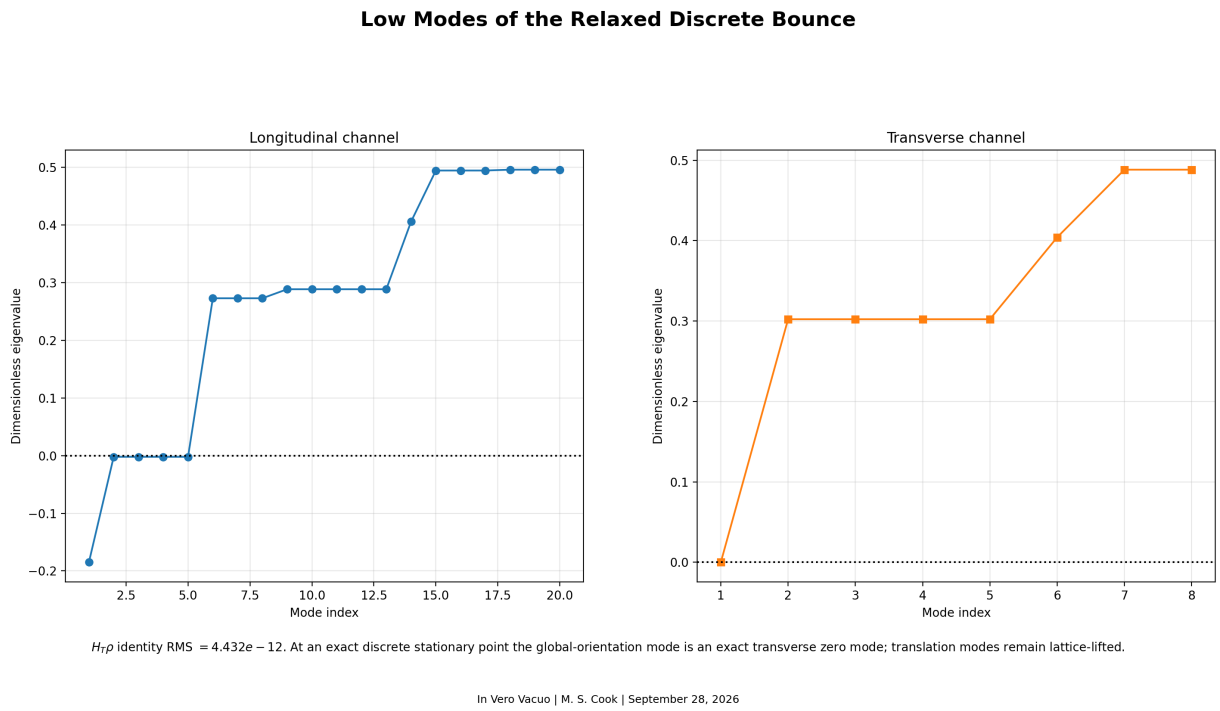


Figure 7.1: Low longitudinal and transverse spectra around the relaxed discrete bounce. The principal longitudinal instability is separated from the four translation modes; the transverse orientation mode is numerically zero.

Table 7.1: Low-mode structure of the relaxed $L = 26$, $h = 0.8$ configuration.

Sector	Multiplicity	Dimensionless eigenvalue
Radial instability	1	$-1.8504151 \times 10^{-1}$
Translations	4	$-2.1265306 \times 10^{-3}$
Internal orientation	3 physical copies	-5.33×10^{-12}
First positive longitudinal level	3-fold	2.7292064×10^{-1}
First positive transverse level	4-fold per channel	3.0226725×10^{-1}

Scope and interpretation

The translation modes are not exactly zero on a fixed lattice because continuous translations are broken to discrete shifts. They are identified by their fourfold degeneracy and collective-coordinate character, then removed from the primed determinant. The radial negative mode is retained through $\log|\lambda_-|$.

7.3 New link-derived SU(2) gauge–Goldstone Hessian

The scalar production pipeline is unchanged, but research mode now provides a separate quadratic gauge extension when `HIGGS_SU2_GAUGE=1`. The background is a radial Higgs doublet with vanishing gauge field. Each of the three generators has the same five-field block $(A_1, A_2, A_3, A_4, \chi)$; the radial Higgs remains a separate scalar block. Thus the bosonic dimension is $16L^4$, or 7,311,616 at $L = 26$, while the three complex ghost operators are handled separately.

7.3.1 Why the link stencil replaces the centered stencil

The initial gauge diagnostic used a centered continuum gradient and a site-local gauge mass. It was self-adjoint, but self-adjointness alone does not imply the discrete gauge identity. The current implementation instead expands the covariant scalar-link and zero-background gauge-link actions. Gauge fields occupy outgoing links, scalar fields occupy sites, and the positive background R_ξ parameter is no longer restricted to $\xi = 1$. Define the forward gradient G, backward divergence D, $G^\dagger = -D$, and $K = -DG$. With $q = g/2$ and $F = DA - \xi q \rho \chi$, the implemented one-generator operators are

$$H_{A\chi} = \begin{pmatrix} K\mathbf{1} + (1-\xi^{-1})GD + M^2_{\mathrm{link}} & C \\ C^\dagger & K + U_T + \xi q^2 \rho^2 \end{pmatrix}, \qquad H_c = K + \xi q^2 \rho^2, \tag{7.4}$$

where

$$(M^2_{\mathrm{link}})_{x\nu} = q^2 \rho_x \rho_{x+\hat{\nu}}, \qquad (C\chi)_{x\nu} = \frac{q}{h}(\rho_{x+\hat{\nu}} - \rho_x)(\chi_x + \chi_{x+\hat{\nu}}). \tag{7.5}$$

The continuum limit of the latter is $2q(\partial_\nu \rho)\chi$. The stored research field is converted to $\rho_{\mathrm{phys}} = \rho/a$ before calling the shared operator constructor. Production validation instead supplies the field and spacing directly in scaled units.

```
for reproducibility, but they are no longer required by this program.
"""

import os as _pipeline_os
import subprocess as _pipeline_subprocess
import sys as _pipeline_sys

from su2_lattice_vertices import (
    faddeev_popov_matrix as make_nonlinear_su2_ghost_matrix,
    generate_one_loop_vertex_tables,
    ghost_gauge_vertex as nonlinear_su2_ghost_gauge_vertex,
    ghost_scalar_vertices as make_nonlinear_su2_scalar_ghost_vertices,
    scalar_link_vertex as nonlinear_su2_scalar_link_vertex,
    wilson_gauge_vertex as nonlinear_su2_wilson_gauge_vertex,
)


def _pipeline_env_flag(name, default=False):
    """Read a conservative Boolean environment flag."""
    value = _pipeline_os.getenv(name)
    if value is None:
        return bool(default)
    normalized = value.strip().lower()
    if normalized in {"1", "true", "yes", "on"}:
        return True
    if normalized in {"0", "false", "no", "off"}:
        return False
    raise ValueError(f"{name} must be one of 0/1, false/true, no/yes, or off/on.")


def make_periodic_scalar_operator(curvature, spacing):
    """Return -Delta plus a site curvature on a periodic four-dimensional grid."""
    import numpy as np
    import scipy.sparse.linalg as spla

    curvature_grid = np.asarray(curvature, dtype=float)
    if curvature_grid.ndim != 4 or not np.all(np.isfinite(curvature_grid)):
        raise ValueError("The scalar curvature must be a finite four-dimensional grid."
            )
    if not np.isfinite(spacing) or spacing <= 0.0:
```

```
        raise ValueError("The scalar-operator spacing must be finite and positive.")

    def matvec(vector):
        scalar = np.asarray(vector, dtype=float).reshape(curvature_grid.shape)
        laplacian = sum(
            (2.0 * scalar - np.roll(scalar, 1, axis=axis)
             - np.roll(scalar, -1, axis=axis)) / spacing**2
            for axis in range(4)
        )
        return (laplacian + curvature_grid * scalar).ravel()

    site_count = curvature_grid.size
    return spla.LinearOperator(
        (site_count, site_count), matvec=matvec, rmatvec=matvec,
        dtype=np.float64
    )


def export_su2_modal_dataset(output_directory, dataset_name, operator,
                             eigenvalues, eigenvectors, *, sector,
                             background, lattice, parameters,
                             explicit_operator=None, degeneracy_tolerance=1.0e-9):
    """Export versioned eigenpairs and metadata for quantum-algorithm benchmarks.

    Eigenvectors are stored as columns with ordinary discrete l2 norm one.  A
    deterministic sign is chosen by making each vector's largest-magnitude
    entry positive.  Degenerate subspaces remain basis-dependent and are
    identified in the JSON sidecar rather than assigned physical vector labels.
    """
    import hashlib
    import json
    from datetime import datetime, timezone
    from pathlib import Path

    import numpy as np
    import scipy.sparse as sp

    output = Path(output_directory)
    output.mkdir(parents=True, exist_ok=True)
```

Listing 7.1: Shared link-action gauge–Goldstone and ghost constructors.

7.3.2 Ward identities, normalization, and determinant scope

For $R\alpha = (G\alpha, q\rho\alpha)$, the off-shell identity contains the scalar EOM residual:

$$H_{A\chi} R\alpha = \begin{pmatrix} \xi^{-1} G(H_c \alpha) \\ q\rho H_c \alpha \end{pmatrix} + \begin{pmatrix} 0 \\ q\alpha[K\rho + V\prime(\rho)] \end{pmatrix}. \tag{7.6}$$

The new validation mode retains this term on arbitrary profiles and checks its smallness on the regenerated stationary bounce. A Gaussian cannot be substituted for the stationary field in a physical prefactor calculation.

The homogeneous test combines $\frac{1}{2}\log\det H_{A\chi}$, $-\log\det H_c$, and the gauge-fixing normalization $\frac{N}{2}\log\xi$. Its measured agreement across three gauges is a finite-volume prerequisite, not a completed bounce-rate calculation. In the research report, coupled and ghost spectra are written to a separate CSV and a third diagnostic PDF page. They are explicitly excluded from the scalar IR determinant and from scalar SLQ/continuum results.

Scope and interpretation

The previous centered-stencil SU(2) Gaussian report is historical diagnostic output. Its numbers must not be reused as results of the revised link operator. Appendix D reproduces the new report and defines the adopted finite EFT matching conditions, including the one-loop-order ρ^{10} and ρ^{12} coefficients.

Chapter 8

Regulated SU(2) Determinants and Counterterms

8.1 Gauge–Goldstone–ghost determinant stage

For one $SU(2)$ generator, the new `su2_determinant` mode evaluates the regulated finite-lattice contribution

$$\Delta\Gamma^{(a)}_{1,g} = \frac{1}{2}\log\frac{\det H_{A\chi,B}}{\det H_{A\chi,F}} - \log\frac{\det H_{c,B}}{\det H_{c,F}}. \tag{8.1}$$

The radial background makes the three generator blocks identical, so the reported gauge-sector logarithm is three times Eq. (8.1). The mode provides an exact eigensystem backend for coupled dimensions no larger than 3000 and a production-oriented backend using low-mode deflation and stochastic Lanczos quadrature. Common Rademacher probes are reused at every value of ξ, reducing the stochastic uncertainty in gauge-parameter differences.

At the symmetric false vacuum, $\rho_F = 0$, each generator contains four constant gauge modes and one constant ghost mode on the periodic box. The output records both raw IR-regulated ratios and primed ratios in which these constant modes are omitted. Removing the constant longitudinal gauge mode also removes a factor from the gauge-fixed Gaussian measure; the implemented relative correction is

$$\Delta\Gamma^{(a)}_{\text{measure}} = \frac{1}{2}\log\xi \tag{8.2}$$

per generator. Raw, raw-normalized, primed, and primed-normalized results are therefore retained separately so that apparent gauge dependence can be audited rather than hidden.

Table 8.1: Exact 3^4 homogeneous determinant cancellation test.

IR mass-squared shift	Spread over $\xi = 0.5, 1, 2$	Status
10^{-8}	1.079×10^{-7}	PASS
10^{-6}	1.079×10^{-5}	PASS
10^{-4}	1.078×10^{-3}	PASS

The decreasing spread is consistent with restoration of the normalized gauge-independent result as the auxiliary IR shift is removed. This is an exact homogeneous validation, not the renormalized production-bounce determinant. The production checkpoint was regenerated and validated on October 2 as described below, but it must be regenerated or transferred into the execution environment before the production scan can begin. Counterterm matching and spacing and volume limits remain necessary afterward.

8.2 October 2 production-checkpoint recovery

The missing $L = 26$, $h = 0.8$ stationary-bounce checkpoint was regenerated from the continuum initial profile using the Newton–MINRES relaxation mode. Three accepted Newton corrections reduced the residual from 3.06017×10^{-3} to the previously archived stationary solution. The recovered field had center value 1.365689301 and was written to

`su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz`.

Before the workspace interruption, the regenerated checkpoint passed the complete prerequisite suite. The measured production-background quantities were

$$B_h = 139.9422784742, \tag{8.3}$$

$$\|\mathcal{E}_h\|_{\mathrm{RMS}} = 4.4316323 \times 10^{-12}, \tag{8.4}$$

$$\|\mathcal{E}_h\|_{\infty} = 6.2693073 \times 10^{-11}. \tag{8.5}$$

The stationary-background on-shell Ward residuals at $\xi = 0.5, 1, 2$ were respectively 1.38×10^{-14}, 2.76×10^{-14}, and 5.52×10^{-14}. The independent homogeneous normalized gauge spread was 1.14×10^{-13}. The resulting validation status was **PASS**, and the surviving machine-readable record is `su2_prefactor_validation_20261002/validation_summary.json`.

Workspace-persistence event

A subsequent cloud-workspace reconnection restored the project without the untracked NPZ directory. Consequently, the production determinant scan did not start and no determinant probes were lost. The checkpoint must be regenerated once more, or copied from persistent external storage, before the scan is launched. The numerical agreement with the archived solution shows that regeneration is deterministic at the reported precision.

8.3 Local EFT basis and pole closure

The `su2_counterterms` mode evaluates the radial restriction of the adopted gauge-invariant local basis,

$$\Delta S_{\mathrm{ct}} = \int \mathrm{d}^4x \Big[\frac{1}{2}\delta Z_{\mathcal{H}} (\partial_\mu \rho)^2 + \frac{1}{2}\delta m^2 \rho^2 + \frac{1}{4}\delta\lambda\rho^4 + \frac{1}{6}\delta\gamma\rho^6 + \frac{1}{8}\delta\delta\rho^8 + \frac{1}{10}\delta c_{10}\rho^{10} + \frac{1}{12}\delta c_{12}\rho^{12} \Big]. \tag{8.6}$$

Each basis integral is stored independently, allowing finite matching coefficients to be updated without recomputing the background invariants.

With $m_F^2 = -\mu^2$, the longitudinal and transverse scalar curvatures are

$$U_L = m_F^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \tag{8.7}$$

$$U_T = m_F^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6. \tag{8.8}$$

The scalar pole polynomial obtained from $U_L^2 + 3U_T^2 - 4m_F^4$ closes through ρ^{12}. In background Feynman gauge, with $q = g/2$, the radial-background pole density including the three gauge generators is

$$P_{\mathrm{total}} = U_L^2 + 3U_T^2 + 6q^2\rho^2 U_T + 9q^4\rho^4 + 24q^2(\partial_\mu\rho)^2. \tag{8.9}$$

Writing $C = (64\pi^2\bar{\epsilon})^{-1}$, the residues that cancel the positive CP_{total} pole for the production parameters and $g = 0.65$ are listed in Table 8.2.

Table 8.2: Derived continuum Feynman-gauge pole residues.

Coefficient	Residue divided by C
$\delta Z_{\mathcal{H}}$	-5.070000
δm^2	18.693000
$\delta\lambda$	-207.8116390625
$\delta\gamma$	338.718000
$\delta\delta$	-82.325600
δc_{10}	-48.640000
δc_{12}	-3.993600

The direct matrix-invariant expression and Eq. (8.9) agree to 3.11×10^{-15}, while the normalized residues cancel the integrated pole density to 1.70×10^{-15}. These are continuum pole residues, not the finite constants converting the implemented lattice regulator to $\overline{\mathrm{MS}}$. The finite entries in `su2_counterterm_coefficients_template.json` therefore remain explicitly unset until the lattice momentum and derivative matching calculations are completed.

Scope and interpretation

The October 1 implementation supplies a regulated determinant architecture, zero-mode and measure policies, the retained local operator basis, and continuum pole closure. It does not yet supply a counterterm-matched, continuum- and volume-extrapolated $SU(2)$ decay prefactor.

Chapter 9

October 5 Single-Core SU(2) Validation and Variance Audits

9.1 Scope, execution policy, and evidence status

The October 5, 2026 work advanced the quadratic $SU(2)$ branch through four stages: a small four-dimensional operator pilot, a homogeneous increased-probe/quadrature audit, a complete-low-mode-multiplet variance comparison, and a nonconstant-background prerequisite audit. This chapter incorporates their test results into the technical guide rather than leaving them only in separate reports. All backgrounds in these October 5 studies were prescribed off-shell diagnostics, not stationary bounces. No production gauged decay prefactor was calculated.

Execution was sequential on one CPU, with CPU affinity {0} and `OPENBLAS_NUM_THREADS`, `OMP_NUM_THREADS`, `MKL_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` set to one. The recorded environment was Python 3.13.15, NumPy 2.4.1, and SciPy 1.17.0. The separate harnesses used definitions from the original source; they did not execute its top-level scalar pipeline or edit the production solver. Its recorded unchanged SHA-256 was

`c592079bdb8c3632f7d0dc413c98c012bff43fbc393931439af4be1b3f9e0b1f`

Result provenance for this revision

The initial pilot summary is present under `su2_L6_single_core_pilot_20261005/`. The increased-probe and multiplet values below are recorded results transcribed from their standalone October 5 reports; the earlier follow-up raw directories and saved bases are absent from the current snapshot. The nonconstant study was a fresh run, and its harness, aggregate audit summary, symbolic closure, and ten counterterm summaries are present. However, its complete raw matrices, spectra, probes, and comparison CSVs are no longer present in this snapshot. This document update checks the available summaries and incorporates the reports; it does not rerun the calculations or claim a new independent verification of missing raw data.

9.2 Four-dimensional $L = 6$ operator pilot

The pilot used a periodic 6^4 lattice with $h = 0.8$, $g = 0.65$, $\xi = 1$, and auxiliary IR mass-squared shift $\epsilon = 0.003$. One generator has 6,480 coupled gauge–Goldstone and 1,296 ghost degrees of freedom. The two profiles were

$$\rho_{\mathrm{hom}}(n) = 0.1, \qquad \rho_{\mathrm{var}}(n) = 0.1 + 0.04 \prod_{\mu=1}^{4} \cos(2\pi n_\mu/6).$$

The varying profile spans 0.06–0.14 and activates gradient mixing. Eight smallest-algebraic eigenpairs were requested in each sector; this fixed count did not certify complete low-mode degeneracies.

Table 9.1: October 5 $L = 6$ direct operator checks. H and V denote homogeneous and varying profiles. Residuals use $\|Hv - \lambda v\|/\max(1, |\lambda|)$ for normalized v.

Case	Sector	Symmetry	Linearity	Minimum λ	Max. residual
H	Coupled	1.17×10^{-17}	1.10×10^{-16}	0.00105625	3.62×10^{-9}
H	Ghost	0	1.07×10^{-16}	0.00105625	9.06×10^{-12}
V	Coupled	1.05×10^{-17}	1.22×10^{-16}	0.00105406	6.58×10^{-10}
V	Ghost	2.55×10^{-17}	1.04×10^{-16}	0.00106681	1.33×10^{-10}

The reported observable throughout these audits is the regulated three-generator gauge-sector combination

$$\begin{aligned}\mathcal{D}_\epsilon = 3\big[&\tfrac{1}{2}(\log\det H_{A\chi,B,\epsilon} - \log\det{}' H_{A\chi,F,\epsilon}) \\ &- (\log\det H_{c,B,\epsilon} - \log\det{}' H_{c,F,\epsilon}) + \tfrac{1}{2}\log\xi\big],\end{aligned} \tag{9.1}$$

where $H_\epsilon = H + \epsilon I$ and B denotes the diagnostic profile. The false-vacuum prime removes four constant gauge modes and one constant ghost mode per generator before shifting. The radial Higgs determinant is not included. Deflated diagnostic modes remain in the determinant through their explicit log eigenvalues; deflation is not this prime policy.

Table 9.2: Initial four-probe SLQ means and one estimated standard error, using integer seed 20261005 and the same probes at each order.

Order	Homogeneous	Varying
10	-29.6992 ± 12.0430	-9.8218 ± 12.7586
20	-45.2381 ± 11.5188	-18.1826 ± 9.9116
30	-45.2406 ± 11.5187	-18.1851 ± 9.9117

The paired mean order-30 minus order-20 changes were -0.002522 and -0.002580; maximum absolute per-probe changes were 0.003087 and 0.003108. Ten steps were insufficient. Stable higher-order estimates at fixed probes did not establish a precise trace. The homogeneous exact Fourier target was $\mathcal{D}_{\epsilon,\text{exact}} = -19.56733155726579$. The initial order-30 mean differed by -25.6733154, or 2.23 estimated standard errors; this discrepancy is retained rather than relabelled as precision validation. The tested operator gate required symmetry and linearity below 10^{-12}, low-pair residuals below 10^{-7}, positive computed minimum eigenvalue plus shift, and finite estimates. Both profiles passed that gate, not a determinant-accuracy gate. The complete pilot took 7.6775 seconds; peak memory was not recorded. A single-ξ automatic gauge-spread PASS is vacuous and is not evidence of gauge independence.

9.3 Increased probes and independent quadrature references

The homogeneous follow-up kept the same lattice, parameters, prime policy, and eight deflation vectors per sector. It used 4,096 independent Rademacher probe pairs with seed 20261005 and orders $10, 20, 30, 40, 60, 80$. Its four-probe prefix is not the initial pilot: a different deflation basis was generated. The 1,024- and 4,096-probe follow-ups reused one saved basis.

The final discrepancy was 0.727 estimated standard errors. The uncertainty approximately halved from 1,024 to 4,096 probes. Only one of the four gauge constant modes was covered by the eight coupled deflation vectors: they were not the complete eight lowest modes counted with multiplicity. The recorded

Table 9.3: Recorded nested prefixes of the homogeneous follow-up at 80 steps. Prefixes are correlated, not independent ensembles.

Probes	Mean	SEM	Mean minus exact
4	−27.734582	18.319092	−8.167251
16	−40.690843	25.294243	−21.123511
64	−26.002996	13.888968	−6.435665
256	−21.893954	6.720888	−2.326622
1024	−20.992059	3.247177	−1.424727
4096	−20.750938	1.628882	−1.183606

Fourier expected-trace bias check was approximately 1.21×10^{-12}, negligible here but not a coverage certificate.

For every probe, an orthonormal Fourier transform evaluated the same projected quadratic form directly, with $P = I - VV^T$:

$$q_{\mathrm{ref}}(z) = \sum_{j \in \mathrm{low}} \log(\lambda_j + \epsilon) + \sum_k |\mathcal{F}(Pz)_k|^2 \log s_k.$$

SLQ minus this individual reference isolates quadrature error rather than sampling error relative to the trace.

Table 9.4: Recorded quadrature error over 4,096 homogeneous probes.

Order	Mean error	Maximum absolute probe error
10	13.6303	66.2927
20	2.5171×10^{-3}	4.4397×10^{-3}
30	-9.18×10^{-12}	6.00×10^{-11}
40	-1.15×10^{-11}	7.09×10^{-11}
60	-1.19×10^{-11}	6.55×10^{-11}
80	-1.14×10^{-11}	1.04×10^{-10}

An independent exact-arithmetic Fourier-measure Gauss/left-Radau bracket used 34 coupled and 17 ghost distinct spectral values. For $\log x$, the sector bounds are $R_m(a) \le \int \log x \, d\mu \le G_m$, with $a = 0.99 \min s_k > 0$. The combined lower and upper bounds use $3(\frac{1}{2}L_c - U_g) + C$ and $3(\frac{1}{2}U_c - L_g) + C$, respectively, where C is the exact false-vacuum contribution. Maximum widths at 10, 20, and 30 steps were 68.5729, 0.26145, and 7.0215×10^{-8}. Finite-support quadrature terminates by 34/17 sector steps in exact arithmetic; the 40–80-step theoretical widths therefore vanish. Numerical reference-bracket violations reached 8.19×10^{-11}. These are not directed-rounding certificates for the production recurrence, and theoretical termination does not imply machine-exact results. This finite-support result does not transfer to a nonconstant background. The recorded 4,096-probe run took 256.1 seconds, including Fourier and bracket audits, and produced 24,576 probe/order rows.

9.4 Complete low-mode multiplets and variance reduction

For the homogeneous control, the audit built real sine/cosine bases for conjugation-closed Fourier masks. The four gauge constant modes, one Goldstone constant mode, and one ghost constant mode can then be covered explicitly. Completing an eight-mode target requires 37/9 coupled/ghost vectors because the boundary intersects degenerate shells. Two independent seeds, 20261005 and 20261006, each used 4,096

probe pairs; the same pairs were reused across six schemes within each seed. These integers are seeds, not execution dates.

For an orthonormal deflation basis, the Rademacher residual estimator has

$$M = P\log(H+\epsilon I)P, \qquad \mathrm{Var}(z^T M z) = 2(\|M\|_F^2 - \sum_i M_{ii}^2).$$

Independent coupled and ghost probes give combined single-probe variance $9(\frac{1}{4}\,\mathrm{Var}_c + \mathrm{Var}_g)$. The homogeneous Fourier structure allowed direct double-precision variance calculation rather than cutoff selection from sampled fluctuations.

Table 9.5: Recorded complete-multiplet comparison. Positive reduction is a variance decrease relative to the freshly generated eight-vector baseline.

Scheme	Coupled/ghost	Probe variance	SEM at 4096	Reduction
No deflation	0/0	11840.37	1.70021	−6.21%
Fresh eight-vector	8/8	11148.31	1.64977	Reference
Complete constants	5/1	9753.25	1.54310	12.51%
Complete eight-mode target	37/9	10327.65	1.58789	7.36%
First shell, all fields	45/9	10430.68	1.59579	6.44%
Second shell, all fields	165/33	14603.28	1.88819	−30.99%

Complete constants were the best tested cutoff: 12.51% less variance, or about 6.47% less standard error at fixed count. This is a modest gain, not a universal optimum. More deflation need not reduce this uncentered estimator variance; the second shell increased the residual off-diagonal log-operator norm. The fresh eight-vector basis again covered only one gauge constant mode and must not be equated with the earlier saved 4,096-probe basis. Complete schemes recorded maximum orthogonality error 4.76×10^{-15} and eigenpair residual 4.59×10^{-16}; the constant scheme expected combined trace-bias check was 8.31×10^{-13}.

Full ensembles used Fourier quadratic forms, not 8,192 full SLQ runs. For complete constants the seed means were -21.266991 ± 1.530133 and -17.952936 ± 1.551980. Pooling only these two independent ensembles yielded

$$\widehat{\mathcal{D}}_\epsilon = -19.609964 \pm 1.089806 \quad \text{(8192 probes; one estimated standard error)}.$$

The difference from exact was -0.042632; uncertainty near 1.09 does not establish four-decimal accuracy. Shared-probe schemes cannot be pooled as independent estimates. Original SLQ was checked on 64 matched pairs per scheme at orders 20, 30, and 40. For complete constants, maximum quadrature errors were 7.55×10^{-9}, 7.64×10^{-11}, and 9.09×10^{-11}; for no deflation they were 1.74, 3.17×10^{-8}, and 7.82×10^{-11}. These subset measurements are not certified full-ensemble bounds. The recorded finalized harness run took approximately 20.3 seconds, dominated by Fourier and subset checks.

9.5 Fresh nonconstant-background prerequisites on $L = 3, 4$

The fresh audit retained $h = 0.8$ and $g = 0.65$ and used periodic 3^4 and 4^4 grids, with coupled/ghost dimensions 405/81 and 1280/256. It swept $\xi = 0.5, 1, 2$ and $\epsilon = 0.001, 0.003, 0.01$ on profiles

$$\rho_A(n) = 0.1 + A\prod_{\mu=1}^{4}\cos(2\pi n_\mu/L).$$

Determinant and SLQ checks used $A = 0, 0.04$; local-basis tests also used $A = 0.005, 0.01, 0.02$. Each determinant case used 64 independent pairs per seed for two seeds, orders 20, 40, and 80, and a target-eight

low window expanded through degeneracy boundaries. Both grids have the same spacing and different physical volumes: this is not a continuum extrapolation.

Explicit matrices were assembled by applying the original matrix-free operators to coordinate unit vectors. Tests used the unsymmetrized matrices; full dense eigensystems supplied deterministic traces and individual projected-probe references. Matrix assembly and operator action share an implementation, so agreement is a consistency check complemented by analytic homogeneous spectra and the off-shell Ward identity.

Table 9.6: Fresh nonconstant-audit maxima, checked against 10^{-10} gates.

Diagnostic	Maximum error
Matrix self-adjointness	0
Linearity	2.04×10^{-16}
Matrix-free versus assembled action	2.59×10^{-16}
Low-eigenpair relative residual	6.28×10^{-14}
Off-shell lattice Ward identity	2.99×10^{-16}
Zero-coupling decoupling at $\xi = 1$	1.48×10^{-16}
Homogeneous versus analytic spectrum	4.97×10^{-14}
Independent local pole-basis closure	1.53×10^{-15}

With $q = g/2$ and $R_\rho = -\Delta\rho + U_T\rho$, the Ward test retained the equation-of-motion term:

$$H_{A\chi}\begin{pmatrix}\nabla\alpha\\ q\rho\alpha\end{pmatrix} = \begin{pmatrix}\nabla(H_c\alpha)/\xi\\ q\rho H_c\alpha + q\alpha R_\rho\end{pmatrix}.$$

Three random α fields were used per gauge parameter. The varying profiles had RMS equation-of-motion residuals 0.194615 ($L = 3$) and 0.133865 ($L = 4$), not approximately zero. Passing this off-shell identity does not establish physical gauge independence.

The low window closes when adjacent levels differ by more than $10^{-9}\max(1, |\lambda|)$. Varying coupled/ghost ranks were 9/8 for all three gauges at $L = 3$, and 17/9, 9/9, 9/9 at $L = 4$ in increasing ξ order. Homogeneous coupled ranks were $29, 37, 13$, with ghost rank nine on both grids. No homogeneous constant-mode shortcut was imposed on nonconstant fields.

Table 9.7: Fresh varying-profile exact determinants and 80-step estimates at $\epsilon = 0.003$. Each seed mean uses 64 pairs; uncertainties are one SEM.

L	ξ	Exact	Seed 20261005	Seed 20261006
3	0.5	-17.4139	-23.4876 ± 5.3701	-15.0475 ± 4.7876
3	1	-16.7945	-24.6894 ± 4.7614	-14.8426 ± 4.4228
3	2	-16.4506	-26.0028 ± 4.7508	-14.3035 ± 4.3497
4	0.5	-17.8121	-9.2855 ± 7.1145	-14.8401 ± 7.4645
4	1	-17.1935	-12.0467 ± 6.6016	-19.8538 ± 6.9532
4	2	-16.8513	-17.4302 ± 7.3510	-21.9011 ± 7.1914

Maximum individual combined SLQ-minus-dense-reference errors across cases and shifts were 1.00×10^{-11} at 20 and 40 steps and 1.05×10^{-11} at 80 steps. Sampling error remained dominant. The largest discrepancy from exact among 72 correlated 80-step ensemble comparisons was 2.01 estimated standard errors; no arbitrary two-SEM pass/fail gate was imposed. Probe reuse across parameters and profiles means those comparisons are correlated.

At $\xi = 1$, the exact varying-profile values were:

L	$\epsilon = 0.001$	$\epsilon = 0.003$	$\epsilon = 0.01$
3	−18.851936	−16.794478	−13.772544
4	−19.247069	−17.193485	−14.173802

The minimum shifted eigenvalue was 0.001528125, so all real logs were defined. The convention check gave $\mathcal{D}_{\text{primed}} - \mathcal{D}_{\text{raw}} = 3\log\epsilon$ at the same gauge-measure normalization. The observed off-shell gauge and IR dependence is retained, not interpreted as a physical cancellation test.

9.6 Local UV-basis checks and finite-matching boundary

The unchanged counterterm stage evaluated ten profiles with no external finite coefficients. Unlike Eq. (9.1), it includes the radial scalar as well as three generator sectors. At $\xi = 1$, an independent sitewise minimal continuum E matrix with $E_{\mu\nu} = m_W^2\delta_{\mu\nu}$, $E_{\chi\chi} = U_T + m_W^2$, and $E_{\mu\chi} = 2q\partial_\mu\rho$ gave the pole density

$$P_{\text{pole}} = U_L^2 + 3\left[2m_W^4 + (U_T + m_W^2)^2 + 8q^2(\partial\rho)^2\right] - 4m_F^4, \qquad m_F^2 = 0.4.$$

A separate symbolic expansion retained potential terms through ρ^{12}. The independent matrix-versus-polynomial error was 1.53×10^{-15}; scalar/total closure and residue-cancellation identities also passed the 10^{-10} diagnostic gate. The gradient coefficient was $24q^2 = 6g^2 = 2.535$ and $\delta Z_H/C = -12g^2 = -5.07$, using the kinetic basis $\frac{1}{2}\int(\partial\rho)^2$ and source pole convention $C = 1/(64\pi^2\epsilon_{\text{bar}})$. The dimensional pole variable ϵ_{bar} is distinct from the auxiliary IR mass-squared shift ϵ.

Forward-difference evaluation of these continuum local expressions obeyed

$$h^4\sum|\nabla\rho_A|^2 = h^2L^4A^2\sin^2(\pi/L).$$

For $A = 0.005, 0.01, 0.02, 0.04$, the gradient integrals were $0.000972, 0.003888, 0.015552, 0.062208$ at $L = 3$ and $0.002048, 0.008192, 0.032768, 0.131072$ at $L = 4$; they vanish at $A = 0$. These tests verify gradient implementation and local pole algebra. They do not measure lattice UV asymptotics, independently derive continuum heat-kernel residues, or determine a finite wave-function constant. Small amplitude on these coarse grids is not evidence of a resolved continuum derivative expansion.

October 5 status and next calculation

The fresh deterministic operator, off-shell Ward, convention, and local pole-algebra checks passed their stated tolerances. The run took 72.23 seconds and recorded 36 determinant references, 13,824 probe/order/shift rows, 24 operator-check rows, and ten local-basis profiles. Quadrature error was negligible relative to sampling uncertainty. All ten counterterm summaries retained `BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED`; full matching remained false and no finite counterterm action was supplied.

Finite matching is the next distinct calculation. It requires an explicit EFT truncation/operator basis, matching scale, gauge convention, and consistent lattice/continuum IR prescription. The finite gradient coefficient must be matched independently, not inferred from its pole residue or fitted blindly to the tiny-grid determinant values. Nonzero gauge backgrounds are additionally required for full gauge-field and coupling matching. A stationary-bounce/collective-mode audit and controlled volume, spacing, and regulator limits remain necessary before interpreting a physical rate.

9.7 October 6 matching-coefficient progress audit

The October 6 morning review audited the coefficient template, the solver's counterterm interface, the symbolic pole closure, the aggregate prerequisite summary, and all ten retained profile summaries. It

also reran Python syntax compilation and the lightweight analytic validation; the latter completed with `VALIDATION RESULT: PASS`. No production-source edit or finite matching calculation was performed during that morning audit.

The completed infrastructure consists of the kinetic and potential basis through ρ^{12}, continuum background-Feynman-gauge pole residues, local closure checks at approximately 10^{-15}, and a JSON interface that validates supplied coefficients and evaluates their action against measured basis integrals. All ten retained nonconstant-profile summaries report `BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED`.

The finite status remains unambiguous:

```
"coefficients": null,    finite_matching_coefficients_derived=false,
                         full_su2_matching_confirmed=false.
```

After the morning audit, a first finite integral was evaluated as a narrow normalization check: the scalar $O(4)$ quartic tadpole at $a = 0.8$, $m^2 = \mu^2 = 0.4$, and $\lambda = -2$. The implemented forward-difference kinetic term and corresponding free kernel are

$$S_{\rm kin}^{\rm lat} = \frac{a^4}{2}\sum_{x,\nu}\left(\frac{\rho_{x+a\hat{\nu}} - \rho_x}{a}\right)^2, \qquad \widehat{k}^{\,2} = \frac{4}{a^2}\sum_{\nu}\sin^2\left(\frac{ak_\nu}{2}\right).$$

For $V_4 = \lambda(\Phi^A\Phi^A)^2/4$ with four real scalar components, the single diagram has $\Sigma_{\rm tad} = 6\lambda I$. The numerical result, using $m^2_{\rm lat} = m^2_{\overline{\rm MS}} + \delta m^2$ as the additive matching convention, is

$$\begin{aligned} I_{\rm lat} &= 0.2268778187404072, \\ I_{\overline{\rm MS}} &= -0.002533029591058444, \\ \delta m^2_{\rm tad} &= 6\lambda(I_{\overline{\rm MS}} - I_{\rm lat}) = 2.752930179977589. \end{aligned}$$

This is only one cutoff-dependent scalar mass contribution. It is not the complete finite mass coefficient, must not be supplied as `delta_m2`, and does not determine the momentum-dependent kinetic coefficient. No complete potential or wave-function matching coefficient was computed. The continuum pole residues listed earlier in this guide are not finite matching constants and must not be substituted for them. The template still has `"coefficients": null`, the full-matching flag remains false, and no finite counterterm action or gauged decay rate follows from this partial example.

The default stationary-bounce checkpoint expected by `su2_counterterms` is absent from the current workspace. An explicitly selected retained diagnostic background can exercise the basis evaluator, but neither route derives the coefficients. The next calculation is an infinite-volume radial matching calculation at $\xi = 1$ and $\mu = \sqrt{0.4}$: derive lattice momentum-space vertices, evaluate regulated Brillouin-zone potential and small-external-momentum amplitudes, subtract the corresponding $\overline{\rm MS}$ amplitudes with a common IR prescription, and verify regulator cancellation and numerical convergence. Bare mass shifts can have power-law cutoff dependence, so test matched amplitudes or observables under spacing refinement rather than requiring each bare coefficient to be a-independent. Keep `includes_full_su2_lattice_to_msbar_matching=false` for any radial-only result. Full gauge-field and gauge-coupling matching additionally requires nonzero background gauge fields.

October 6 matching status

The achieved milestone is basis and pole-residue validation plus a working coefficient-input interface, with one scalar tadpole contribution evaluated as a first matching check. The complete finite lattice-to-$\overline{\rm MS}$ coefficient calculation remains incomplete. Do not present the tadpole alone as a derived mass coefficient or full matching result.

9.7.1 October 7–8 nonlinear action, routing, and matching projections

The follow-up procedure report makes the target theory explicit while preserving the implementation boundary: the production fluctuation and determinant operators still use the scalar forward-difference action and quadratic gauge–Goldstone–ghost operators. The nonlinear vertex generators are implemented in `su2_lattice_vertices.py`, imported by the solver, and available through its table-generation mode; they are not yet consumed by a full-gauged matching integrator. For full-theory matching, the proposed convention is a fundamental Higgs doublet with links $U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a]$, $T^a = \sigma^a/2$, and the Wilson plaquette action

$$S_g = \frac{4}{g_0^2} \sum_{x,\mu<\nu} \left(1 - \frac{1}{2}\,\mathrm{ReTr}\, U_{x,\mu\nu}\right), \qquad U_{x,\mu\nu} = U_{x\mu} U_{x+a\hat{\mu},\nu} U^\dagger_{x+a\hat{\nu},\mu} U^\dagger_{x\nu}.$$

With this generator and link normalization its continuum limit is $\frac{1}{4}\int d^4x\, F^a_{\mu\nu} F^a_{\mu\nu}$, the Wilson plaquette formulation [14]. The gauge coupling remains the bare g_0; the benchmark $g(\mu) = 0.65$ alone does not specify $g_0(a)$.

The matter action is the covariant nearest-neighbor term

$$S_{H,\mathrm{link}} = a^4 \sum_{x,\mu} \frac{(U_{x\mu} H_{x+a\hat{\mu}} - H_x)^\dagger (U_{x\mu} H_{x+a\hat{\mu}} - H_x)}{a^2}, \qquad \rho_x = \sqrt{2 H_x^\dagger H_x}.$$

At $U = 1$ it reduces to the canonical four-real-component forward difference already used by the solver. The tree potential remains $V_0 = -\mu_{\mathrm{sq}}\rho^2/2 + \lambda\rho^4/4 + \gamma\rho^6/6 + \delta\rho^8/8$; the ρ^{10} and ρ^{12} terms belong to the perturbative matching-counterterm basis, not the leading tree potential. The gauge-covariant kinetic counterterm is δZ_H times $S_{H,\mathrm{link}}$.

For a radial background and zero background gauge field, the proposed background gauge is

$$F^a_x = (DA^a)_x - \xi q \bar{\rho}_x \chi^a_x, \qquad S_{\mathrm{gf}} = \frac{a^4}{2\xi} \sum_{x,a} (F^a_x)^2, \qquad q = g_0/2,$$

with the Faddeev–Popov operator defined as the exact variation of F under the stated lattice gauge transformations. This defines the nonlinear ghost rules consistently with the link-field coordinate; it is not part of the gauge-invariant bare action. The proposed perturbative action is $S_g + S_{H,\mathrm{link}} + a^4 \sum_x (V_0 + V_{\mathrm{ct}}) + S_{\mathrm{gf}} + S_{\mathrm{gh}}$. Its quadratic matter-link expansion agrees with the recorded link Hessian, but that check does not validate the missing nonlinear vertices.

The procedure report specifies exact vertex generators rather than a continuum derivative approximation: expand the two exponentials in each scalar link to generate scalar–scalar–gauge vertices (including the terms linear and quadratic in A), differentiate the ordered plaquette product for pure-gauge vertices, and differentiate $M = \delta F/\delta\omega$ for ghost vertices. Scalar-potential vertices follow by differentiating $f(X)$ with $X = \phi^A\phi^A$; the compact derivative rule and explicit link expansions are given in the procedure report. This is the standard perturbative lattice-vertex construction [15].

At one loop, for constant ρ, $\xi = 1$, and $M_A^2 = q^2\rho^2$, define

$$U_L = m_0^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \qquad U_T = m_0^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6.$$

The lattice determinant contribution is

$$V_{1,\mathrm{lat}}(\rho) = \frac{1}{2}\int_{\mathrm{BZ}} \frac{d^4k}{(2\pi)^4} \left[\log(\hat{k}^2 + U_L) + 3\log(\hat{k}^2 + U_T + M_A^2) + 6\log(\hat{k}^2 + M_A^2)\right],$$

up to a background-independent constant. For the continuum comparison in $d = 4 - 2\epsilon$, the vector/ghost logarithm carries multiplicity $3(d-2)$ before $\overline{\mathrm{MS}}$ subtraction; replacing it by its four-dimensional value

too early can lose a finite term. This determinant is a background-field check, not by itself the full set of local matching coefficients [16, 17].

The kinetic coefficient is defined by the difference of off-shell self-energy slopes at the subtraction point, with lattice $\widehat{p}^2$ and continuum p^2:

$$\delta Z_H(\mu) = \left.\frac{\partial \Sigma^{(2)}_{\overline{\mathrm{MS}},\mathrm{ren}}}{\partial p^2}\right|_{p^2=\mu^2} - \left.\frac{\partial \Sigma^{(2)}_{\mathrm{lat}}}{\partial \widehat{p}^2}\right|_{\widehat{p}^2=\mu^2}.$$

Potential coefficients require matching corresponding multi-point 1PI vertices with an operator projector that separates local potential terms from momentum-dependent operators. Use a common infrared prescription and identical external kinematics on both sides. In particular, the massless gauge contribution contains nonanalytic $\rho^4 \log \rho^2$ behavior, so a Taylor expansion of the constant-background determinant at $\rho = 0$ is not a valid full-gauge-theory potential projector. Nonexceptional Euclidean momenta at a declared subtraction scale, or a common explicit regulator with demonstrated cancellation, are needed before extracting local coefficients.

October 8 full-gauged Step 1 status: no full matching result yet

The October 8 routing/projector deck and its independent tree-level audit complete the selected kinematic setup for full-gauged Step 1. The deck fixes the nonexceptional routings, stores complete generated color/polarization tensors at those selected momenta, and records projectors and a common infrared-regulator sequence. A separate pilot now evaluates the gauge-linked four-radial-Higgs insertion sum at the fixed local-contact routes. It is one projected sector only: it neither supplies the complete four-point amplitude nor establishes regulator cancellation for the full gauge/ghost system. The coefficient JSON must remain null and `includes_full_su2_lattice_to_msbar_matching` must remain false. The scalar quartic tadpole reported above is only one scalar mass-diagram check; it cannot be promoted to the complete `delta_m2` or used to infer `delta_ZH`. Likewise, the gauge-linked four-point diagnostic is not the full `delta_lambda`.

9.7.2 October 8 fixed full-gauged kinematics and projectors

The fixed deck is generated by `su2_full_gauged_projection_setup.py`; its full-precision arrays and metadata are `su2_full_gauged_projection_tables_20261008.npz` and `su2_full_gauged_projection_tables_20261008.json`. The setup uses $a = 0.8$, $g_0 = 0.65$, $\xi = 1$, $\mu_R^2 = 0.4$, and $\bar{\rho} = 0.37$ for local vertex tests. All legs are incoming; site fields use $e^{+ip\cdot x}$ and gauge fields are assigned to link midpoints.

For the Higgs two-point slope, choose $p = \sqrt{s}(1,2,3,4)/\sqrt{30}$ at $s = p^2 = 0.2, 0.4, 0.6$, with legs $(p, -p)$. Fit the three samples to $c_0 + c_1 x + c_2 x^2$, where $x = p^2$ in the continuum and $x = \widehat{p}^2$ on the lattice. The derivative row $(0, 1, 2x_\mu)$, applied to the inverse design matrix, defines the slope at $p^2 = \mu_R^2$ or its corresponding lattice $\widehat{p}^2$; at the subtraction point the latter is 0.3966567139724254. Both continuum and lattice tree kernels return unit slope within 1.11×10^{-15}. This is only a tree-projector test; applying it to $\Sigma^{(2)}$ remains Step 2.

The deck also includes an asymmetric nonexceptional $H^\dagger H A$ triangle, a symmetric three-leg triangle and four-leg tetrahedron at the reference scale, and fixed-shape nonexceptional four-, six-, eight-, ten-, and twelve-scalar contact routings at average squared-momentum scales $0.2, 0.4, 0.6$. For the gauge two-point function define $K = p$ in the continuum and $K_\mu = 2\sin(ap_\mu/2)/a$ on the lattice; use

$$T_{\mu\nu} = K^2\delta_{\mu\nu} - K_\mu K_\nu, \qquad L_{\mu\nu} = K_\mu K_\nu, \qquad P_T^{\mu\nu} = \frac{T^{\mu\nu}}{3(K^2)^2}, \quad P_L^{\mu\nu} = \frac{L^{\mu\nu}}{(K^2)^2}.$$

For each local scalar $2n$-point contact, contract the full tensor and fit the three routing samples to $d_0 + d_1\bar{s} + d_2\bar{s}^2$; retain the intercept d_0. The lattice and continuum fits use their respective $\widehat{p}^2$ and p^2 coordinates.

The common Step 2 infrared prescription is background Feynman gauge and the same additive regulator on every massless gauge and ghost denominator, $k^2 \to k^2 + \lambda_{\mathrm{IR}}^2$ and $\widehat{k}^{\,2} \to \widehat{k}^{\,2} + \lambda_{\mathrm{IR}}^2$, at the deck's base points $\lambda_{\mathrm{IR}}/\mu_R = 1/8, 1/16, 1/32$. Massive scalar lines remain unshifted. The gauge-linked four-point pilot applies this prescription at those points and adds $1/64$ as a check on its three-point extrapolation. This is not yet a cancellation test for the complete gauge/ghost system: regulator cancellation and vanishing of regulator-induced Ward-identity breaking must still be demonstrated in the full $\lambda_{\mathrm{IR}} \to 0$ extrapolation.

The independent audit `su2_full_gauged_projection_audit.py` records PASS in `su2_full_gauged_projection_audit_20261008.json`. Its selected checks include momentum conservation and nonexceptionality, scalar-link tensors against finite-action differences, the asymmetric scalar-link Ward identity, Wilson tensors against an independent expansion and two-point transversality, ghost tensors against finite transformations, the nonzero-background Faddeev–Popov Jacobian, and tree/local projector algebra. Maximum discrepancies range from 4.65×10^{-16} for momentum conservation to 2.72×10^{-6} for the finite-transformation ghost-tensor comparison; each passes its declared tolerance. These are selected tree-level implementation tests, not loop Ward identities or matching results.

9.7.3 October 8 gauge-linked four-point pilot

The first integrated gauge-linked multi-point sector is a fixed-route four-radial-Higgs projection. It is evaluated by the standalone `su2_four_point_gauge_sector_20261008.py`; its full-precision record is `su2_four_point_gauge_sector_20261008_ircheck.json`. The evaluator expands the one-loop bosonic trace-log through four source derivatives and keeps only the gauge-linked insertion classes. The scalar-only $O(4)$ loops are not included in this sector result; the direct Higgs–ghost contribution vanishes at the symmetric background in the stated gauge.

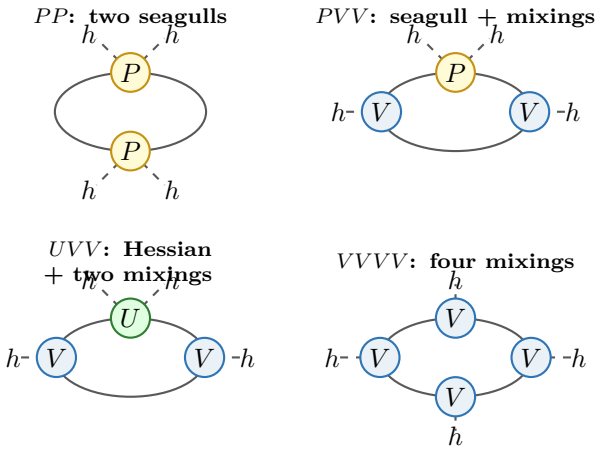


Figure 9.1: Closed-loop insertion classes retained in the gauge-linked four-Higgs pilot. P is the $hhAA$ seagull and carries two external h legs; $V = hA\chi$ is the scalar-link mixing and carries one; U is the λh^2 Goldstone-Hessian insertion and carries two. The loop lines represent the coupled gauge/Goldstone propagators; at each V insertion the internal species changes between gauge and Goldstone. Scalar-only loops are excluded from this pilot.

The continuum integrand is renormalized by subtracting

$$\frac{C}{(k^2+\mu_R^2)^2}, \qquad C = -54q^4 + 36q^2\lambda = -8.20745859375, \qquad q = \frac{g_0}{2}.$$

At the chosen scale this subtraction has zero renormalized $\overline{\mathrm{MS}}$ integral. Lattice and continuum amplitudes use the same three fixed external momentum shapes and the same additive regulator on every internal

gauge line. Figure 9.2 shows the projected continuum-minus-lattice result as the regulator is reduced.

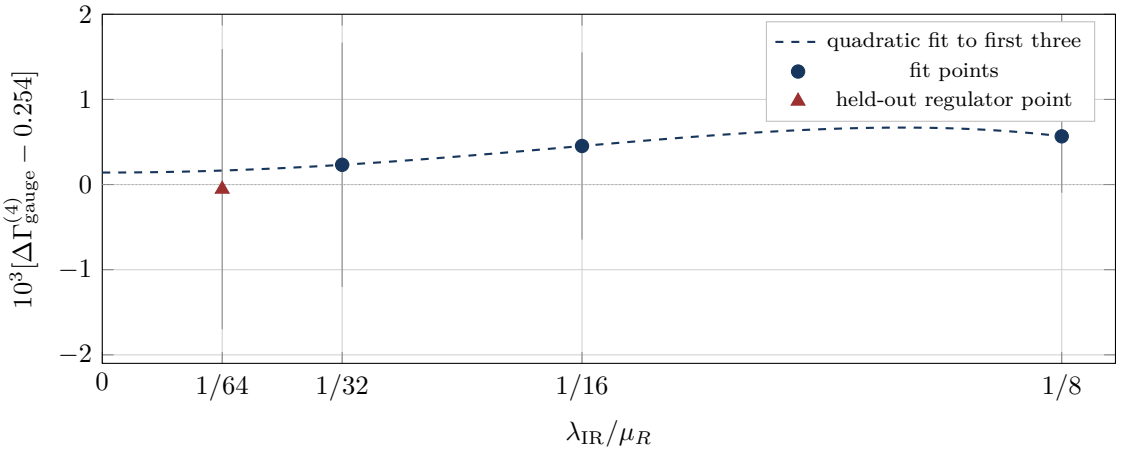


Figure 9.2: Four-point gauge-linked continuum-minus-lattice result versus IR regulator. Error bars are replicate QMC standard errors only. The dashed curve is quadratic in $(\lambda_{\text{IR}}/\mu_R)^2$, fitted to the three blue points; the red triangle is a held-out check, not part of the fit. The zero horizontal line marks the reference value $\Delta\Gamma^{(4)} = 0.254$.

For $\lambda_{\text{IR}}/\mu_R = 1/8, 1/16, 1/32, 1/64$, respectively, $\Delta\Gamma^{(4)}_{\text{gauge}}$ is $0.2545663, 0.2544528, 0.2542318, 0.2539455$; the corresponding QMC standard errors are $6.64 \times 10^{-4}, 1.10 \times 10^{-3}, 1.43 \times 10^{-3}, 1.65 \times 10^{-3}$. The fit to the first three points gives the intercept 0.2541409 ± 0.0015671 (QMC error only). The fourth point's residual from the fit is $-2.19 \times 10^{-4} \pm 1.56 \times 10^{-4}$, about 1.4 QMC standard errors. This is a consistency check, not a robust bound on the extrapolation-model error. Dividing by 3! gives the diagnostic $+0.04236 \pm 0.00026$ for this gauge-linked local-contact projection; it is not the full $\delta\lambda$ and is not written to the coefficient template.

The insertion evaluator reproduces the constant-source gauge/Goldstone determinant to below 7×10^{-13}, agrees with the exact link-generated $hA\chi$ and $hhAA$ vertices to below 6×10^{-17}, and reproduces the large-momentum C/k^4 tail to relative error 8.1×10^{-7}. Its projected integrand is Bose symmetric to numerical precision. These checks validate the selected insertion sum and numerical implementation; they do not constitute a full gauge/ghost Ward or Slavnov–Taylor test.

9.8 Companion reports and retained artifacts

The detailed standalone presentations are:

- `SU2_L6_Single_Core_Pilot_Narrative_Report_20261005.tex`;
- `SU2_L6_Complete_Multiplet_Report_20261005.tex`;
- `SU2_Nonconstant_Prerequisite_Report_20261005.tex`;
- `SU2_Matching_Coefficient_Progress_Report_20261006.tex`;
- `SU2_Next_Steps_Finite_Matching_20261006.tex` (updated October 8; nonlinear action, fixed full-gauged routing/projector deck, independent audit, scalar tadpole check, and gauge-linked four-Higgs pilot).

The original pilot summary is `su2_L6_single_core_pilot_20261005/pilot_summary.json`. Historical follow-up directories `su2_L6_convergence_20261005/`, `su2_L6_convergence_4096_20261005/`, and

`su2_L6_complete_multiplets_20261005/` are not present here; their saved bases and raw probes must be restored to reproduce those basis-dependent comparisons.

The fresh harness `su2_nonconstant_prerequisite_audit.py` is retained.

Within `su2_nonconstant_prerequisites_20261005/`, the retained files include `audit_summary.json`, `symbolic_pole_closure.json`, the archive README, and ten per-profile counterterm summaries. The recorded full archive additionally contained background arrays, sparse operators, complete spectra, low eigenpairs, saved probes, CSVs, and logs. Those raw artifacts and the previously generated bundle are absent from the current snapshot; aggregate summaries are not a substitute. No production bounce or determinant rerun was performed for this guide revision; the separate October 8 vertex/projector audit is recorded below.

Chapter 10

Low-Mode Extraction and the IR Determinant

10.1 ARPACK and degeneracy-safe channel solves

The code applies `scipy.sparse.linalg.eigsh` separately to H_L and H_T. Separate scalar-channel solves reduce memory and preserve the exact threefold transverse degeneracy analytically. The false-vacuum spectrum is constructed from periodic Fourier levels rather than requesting an arbitrary cut through a highly degenerate ARPACK multiplet.

Mode vectors are normalized and diagnosed using their peak amplitude, inverse-participation ratio, and longitudinal/transverse component weights. These quantities separate localized bounce fluctuations from extended periodic plane-wave states.

10.2 Finite spectral-window determinant

For the new homogeneous SU(2) reference operators, the program also avoids ARPACK and retains complete momentum multiplicities analytically. With $k = \widehat{p}^2$ and constant $m_W^2 = q^2\rho^2$, one generator has three levels $k + m_W^2$, one $k/\xi + m_W^2$, and one $k + U_T + \xi m_W^2$ per momentum; its ghost level is $k + \xi m_W^2$. These are sorted without dropping degenerate members. The symmetric false vacuum includes exactly zero gauge and ghost levels. Their gauge-volume prescription remains necessary before forming a physical determinant ratio.

For a retained set of modes,

$$\log\left|\frac{\det{}' H_B}{\det H_F}\right|_{\mathrm{IR}} = \sum_{i\in B'} \log|\lambda_i^{(B)}| - \sum_{j\in F} \log \lambda_j^{(F)}. \tag{10.1}$$

This IR window is a useful diagnostic, but it is not the full determinant because the ultraviolet spectrum is absent. The source therefore reports it explicitly as an IR quantity rather than silently interpreting it as the physical prefactor.

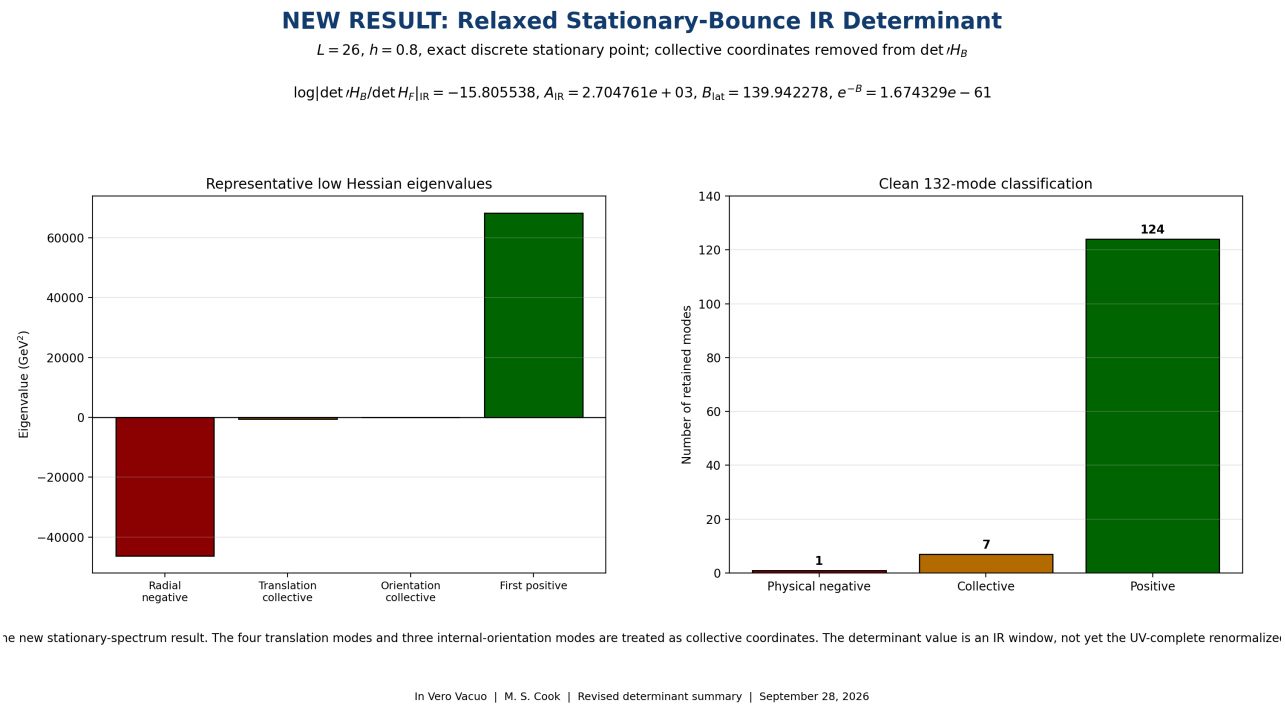


Figure 10.1: Summary of the relaxed stationary spectrum and its primed IR determinant window. This intermediate result motivated the subsequent full SLQ trace–log calculation.

Chapter 11

Deflated Stochastic Lanczos Quadrature

11.1 Trace–log formulation

The determinant is converted into a trace,

$$\log\det H = \operatorname{Tr}\log H. \tag{11.1}$$

For independent Rademacher vectors z_k with entries ± 1,

$$\operatorname{Tr} f(H) \approx \frac{1}{N_p}\sum_{k=1}^{N_p} z_k^T f(H) z_k. \tag{11.2}$$

Lanczos iteration replaces each large quadratic form by Gaussian quadrature on a small tridiagonal matrix T_m,

$$z^T f(H) z \approx \|z\|^2 e_1^T f(T_m) e_1. \tag{11.3}$$

The Rademacher trace estimator and its combination with Lanczos quadrature are described in Refs. [5, 9]. The same random probe is used in bounce and false-vacuum channels, which correlates the estimates and reduces noise in their difference.

11.2 Low-mode deflation

The logarithm varies rapidly near zero. The lowest eigenpairs are therefore calculated exactly, projected out of every random probe, and restored analytically. This both protects collective-coordinate classification and improves Krylov convergence.

```

        if modal_export:
            xi_label = str(gauge_xi).replace(".", "p")
            if method == "exact":
                coupled_export_values = coupled_values[:min(modal_modes, coupled_values
                    .size)]
                coupled_export_vectors = coupled_vectors[:, :coupled_export_values.size
                    ]
                ghost_export_values = ghost_values[:min(modal_modes, ghost_values.size)
                    ]
                ghost_export_vectors = ghost_vectors[:, :ghost_export_values.size]
                coupled_explicit = (coupled_matrix if coupled.shape[0]
                                    <= modal_max_operator_dim else None)
                ghost_explicit = (ghost_matrix if ghost.shape[0]
                                  <= modal_max_operator_dim else None)
            else:
```

```
                coupled_export_values = coupled_computed[:modal_modes]
                coupled_export_vectors = coupled_vectors[:, :coupled_export_values.size
                    ]
                ghost_export_values = ghost_computed[:modal_modes]
                ghost_export_vectors = ghost_vectors[:, :ghost_export_values.size]
                coupled_explicit = None
                ghost_explicit = None
            common_background = {
                "type": "relaxed_bounce", "field_array": bounce,
                "checkpoint": str(bounce_file)
            }
            common_lattice = {
                "shape": list(bounce.shape), "spacing": spacing,
                "boundary_conditions": "periodic"
            }
            metadata_path, _, _ = export_su2_modal_dataset(
                modal_output,
                f"bounce_L{bounce.shape[0]}_xi_{xi_label}_coupled_A_chi",
                coupled, coupled_export_values, coupled_export_vectors,
                sector="coupled_A_chi", background=common_background,
                lattice=common_lattice,
                parameters={"g": gauge_coupling, "xi": gauge_xi,
                            "ir_shift": 0.0, "generator_multiplicity": 3},
                explicit_operator=coupled_explicit
            )
            modal_datasets.append(str(metadata_path))
            metadata_path, _, _ = export_su2_modal_dataset(
                modal_output, f"bounce_L{bounce.shape[0]}_xi_{xi_label}_ghost",
                ghost, ghost_export_values, ghost_export_vectors,
                sector="ghost", background=common_background,
                lattice=common_lattice,
                parameters={"g": gauge_coupling, "xi": gauge_xi,
                            "ir_shift": 0.0, "generator_multiplicity": 3,
                            "statistics": "complex_Grassmann"},
```

Listing 11.1: Matrix-free SLQ operator, low-mode cache, Lanczos recurrence, and quadrature.

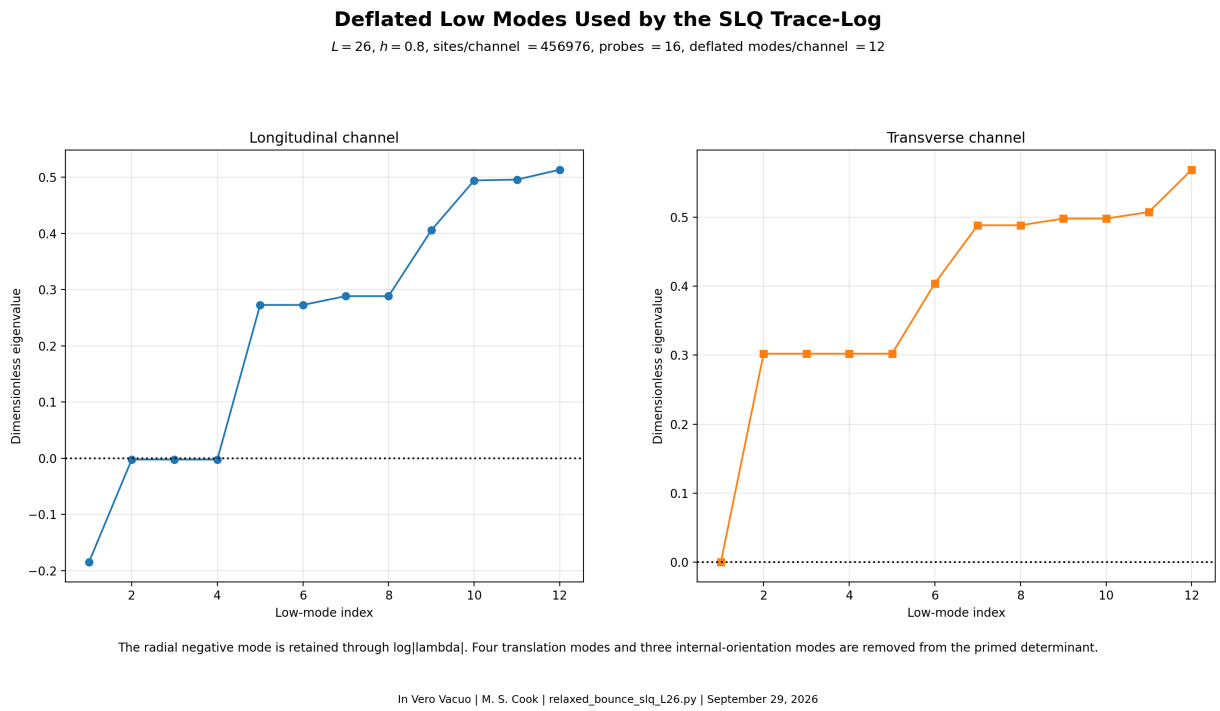
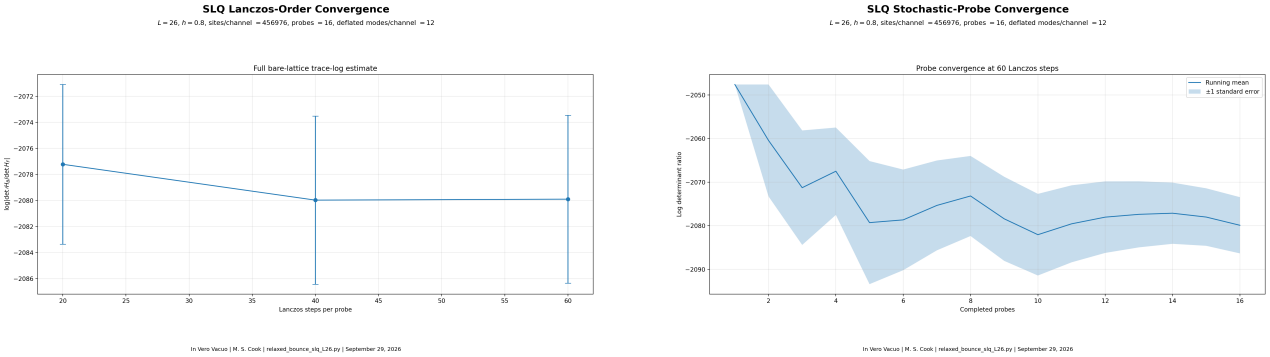


Figure 11.1: Deflated longitudinal and transverse low modes. Four translation modes and three orientation modes are removed; the physical negative mode is retained through its absolute value.



(a) Lanczos-order convergence.

(b) Running stochastic-probe convergence.

Figure 11.2: Two convergence tests required for the stochastic trace–log: Krylov order and probe count.

Table 11.1: Production SLQ settings and result.

Quantity	Value
Lattice	26^4, $h = 0.8$
Sites per channel	456,976
Rademacher probes	16
Deflated eigenpairs per bounce channel	12
Lanczos orders tested	20, 40, 60
Bare finite-lattice trace–log	-2079.9042 ± 6.4504
$\log_{10}$ of bare determinant factor	451.6454 ± 1.4007

Scope and interpretation

The enormous bare factor is not a physical enhancement. It is dominated by the ultraviolet regulator and diverges as the lattice spacing is removed. The result is valuable as a complete finite-lattice diagnostic, but it must be matched to a renormalized continuum prescription.

Chapter 12

Continuum Partial-Wave Renormalization

12.1 Radial decomposition

For an $O(4)$-symmetric background, each scalar operator decomposes into radial partial waves labeled by $\ell = 0, 1, 2, \ldots$ with degeneracy

$$g_\ell = (\ell + 1)^2. \tag{12.1}$$

The radial operator has the form

$$\mathcal{H}_{\ell,c} = -\frac{\mathrm{d}^2}{\mathrm{d}r^2} - \frac{3}{r}\frac{\mathrm{d}}{\mathrm{d}r} + \frac{\ell(\ell+2)}{r^2} + U_c(r), \qquad c \in \{L, T, F\}. \tag{12.2}$$

Gel'fand–Yaglom ratios replace functional determinants by ratios of radial initial-value solutions [4]. Their radial partial-wave extension and renormalized angular-momentum summation are discussed in Ref. [8]. The code uses a stable logarithmic formulation and subtracts the large-ℓ asymptotic terms before summation.

```
        longitudinal_curvature**2
        + 3.0 * (
            2.0 * gauge_mass_sq**2
            + (transverse_curvature + gauge_mass_sq)**2
            + 8.0 * charge_sq * np.sum(gradient**2, axis=0)
        )
        - 4.0 * positive_mass_sq**2
    )
    direct_total_integral = float(cell_volume * np.sum(direct_total_density))
    expanded_total_integral = float(
        sum(
            coefficient * cell_volume * np.sum(bounce**powers[name])
            for name, coefficient in total_closure_polynomial.items()
        )
        + gauge_kinetic_coefficient * cell_volume * np.sum(gradient**2)
    )
    total_closure_relative_error = abs(
        direct_total_integral - expanded_total_integral
    ) / max(1.0, abs(direct_total_integral))
    total_residue_action_over_C = float(sum(
        total_residues_over_C[name] * basis_integrals[name]
        for name in total_residues_over_C
    ))
    total_residue_cancellation_error = abs(
        total_residue_action_over_C + direct_total_integral
```

```
    ) / max(1.0, abs(direct_total_integral))
    if max(total_closure_relative_error, total_residue_cancellation_error) > 1e-11:
        raise RuntimeError("The Feynman-gauge SU(2) pole-residue identity failed.")

    supplied_coefficients = None
    coefficient_metadata = {}
    counterterm_action = None
    full_su2_matching = False
    if coefficient_file_name:
        coefficient_file = Path(coefficient_file_name)
        with open(coefficient_file, encoding="utf-8") as stream:
            coefficient_document = json.load(stream)
        if "coefficients" in coefficient_document:
            supplied_coefficients = coefficient_document["coefficients"]
            coefficient_metadata = coefficient_document.get("metadata", {})
        else:
            supplied_coefficients = coefficient_document
        if supplied_coefficients is not None:
            allowed = set(basis_integrals)
            unknown = set(supplied_coefficients) - allowed
            if unknown:
                raise ValueError(f"Unknown counterterm coefficients: {sorted(unknown)}"
                    )
            supplied_coefficients = {
                name: float(supplied_coefficients.get(name, 0.0)) for name in allowed
            }
            if not all(np.isfinite(list(supplied_coefficients.values()))):
                raise ValueError("All supplied counterterm coefficients must be finite.
                    ")
            counterterm_action = float(sum(
                supplied_coefficients[name] * integral
                for name, integral in basis_integrals.items()
            ))
        full_su2_matching = bool(coefficient_metadata.get(
            "includes_full_su2_lattice_to_msbar_matching", False
        ))

    status = (
        "FULL_SU2_MATCHING_EVALUATED" if full_su2_matching
        else "COEFFICIENTS_EVALUATED_NOT_FULL_SU2" if supplied_coefficients is not None
        else "BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED"
    )
    basis_rows = []
    for name, integral in basis_integrals.items():
        basis_rows.append({
            "coefficient": name, "basis_integral": integral,
            "supplied_value": (supplied_coefficients or {}).get(name),
            "action_contribution": (
                (supplied_coefficients or {}).get(name, 0.0) * integral
                if supplied_coefficients is not None else None
            ),
            "scalar_residue_over_C": scalar_residues_over_C.get(name),
            "feynman_gauge_total_residue_over_C": total_residues_over_C.get(name),
        })
    basis_file = output / "counterterm_basis.csv"
    with open(basis_file, "w", newline="", encoding="utf-8") as stream:
        writer = csv.DictWriter(stream, fieldnames=list(basis_rows[0]))
        writer.writeheader()
        writer.writerows(basis_rows)

    summary = {
```

```
        "status": status,
        "bounce_checkpoint": str(bounce_file), "shape": list(bounce.shape),
        "spacing": spacing, "matching_scheme": "MSbar",
        "matching_scale": matching_scale,
        "benchmark_parameters": {
            "m2": positive_mass_sq, "lambda": lambda_value,
            "gamma": gamma, "delta": eighth, "g": gauge_coupling,
            "c10_at_matching_scale": 0.0, "c12_at_matching_scale": 0.0,
            "ZH_at_matching_scale": 1.0,
        },
        "basis_integrals": basis_integrals,
        "scalar_closure_polynomial": closure_polynomial,
        "scalar_closure_relative_error": closure_relative_error,
        "scalar_counterterm_residues_over_C": scalar_residues_over_C,
```

Listing 12.1: Channel potentials, radial Gel'fand–Yaglom ratio, and asymptotic partial-wave subtraction.

12.2 Primed zero-mode limit

The translation modes occur in the longitudinal $\ell = 1$ sector, with multiplicity four. The global-orientation modes occur in the transverse $\ell = 0$ sector, with three physical copies. A small positive spectral shift q regulates these channels. After explicitly dividing by the appropriate powers of q, the calculation is extrapolated to $q \to 0$. This regulated primed-determinant construction is closely related to standard zero-mode determinant treatments [6, 7].

12.3 Renormalization result

This section retains the previously reported global-scalar calculation. The partial-wave, zero-mode, and renormalization strategy follows the established numerical one-loop bounce literature [7]. The quoted ± 0.0300 is hardcoded as an estimated uncertainty in the source summary, rather than derived by an automated convergence analysis. It must not be treated as a measured uncertainty or applied to the gauged calculation. No lattice-to-continuum gauge matching has been implemented.

At the scale $\mu = m_F = \sqrt{0.4}$, with radial cutoff 20 and $\ell_{\max} = 70$, the calculation gives

Key result

$$\log \left| \frac{\det' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}} = 40.9324 \pm 0.0300,$$
$$\left| \frac{\det' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}}^{-1/2} = 1.2931 \times 10^{-9}.$$

The bare value -2079.9 and the finite value 40.93 are not competing answers: they refer to different regulator treatments. The second is the renormalized global-scalar determinant.

Table 12.1: Bare and renormalized determinant quantities.

Quantity	Value	Interpretation
Bare lattice trace–log	-2079.904 ± 6.450	Regulator dependent
Renormalized trace–log	40.9324 ± 0.0300	Finite global-scalar result
Renormalized factor	1.2931×10^{-9}	One-loop suppression
Renormalization scale	$\mu = m_F = \sqrt{0.4}$	Stated scheme choice
Bounce action	145.203947	Controls e^{-B}

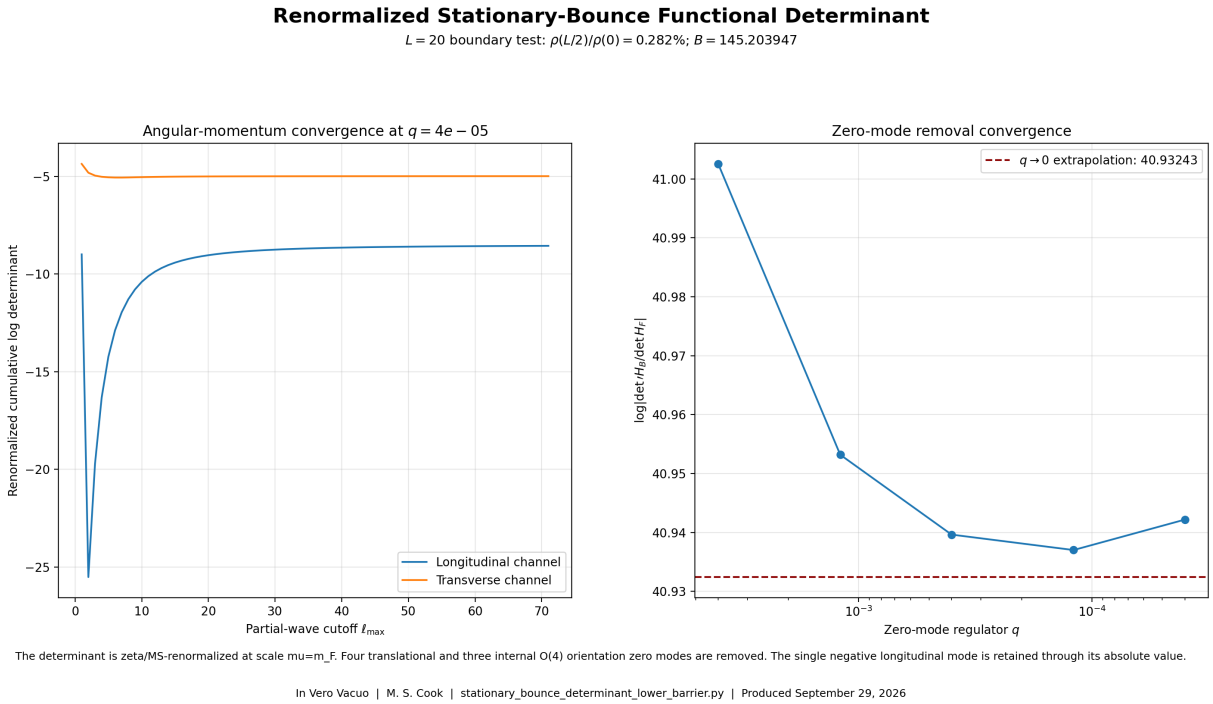


Figure 12.1: Continuum determinant convergence. The panels show the regulated zero-mode limit, partial-wave behavior, and the resulting finite primed determinant.

Chapter 13

Spacing Refinement and Regulator Diagnostics

The continuum bounce sampled on a coarse lattice produced lifted translation and orientation modes. A spacing-refinement study separated finite-volume effects from discretization effects and motivated the direct discrete-action relaxation. The sequence was:

1. verify that the bounce tail fits comfortably inside the box;
2. reduce the spacing while holding the physical box sufficiently large;
3. observe the collective-mode shifts move toward zero;
4. solve the discrete equation directly at the selected refinement;
5. use the relaxed configuration for the Hessian and SLQ calculations.

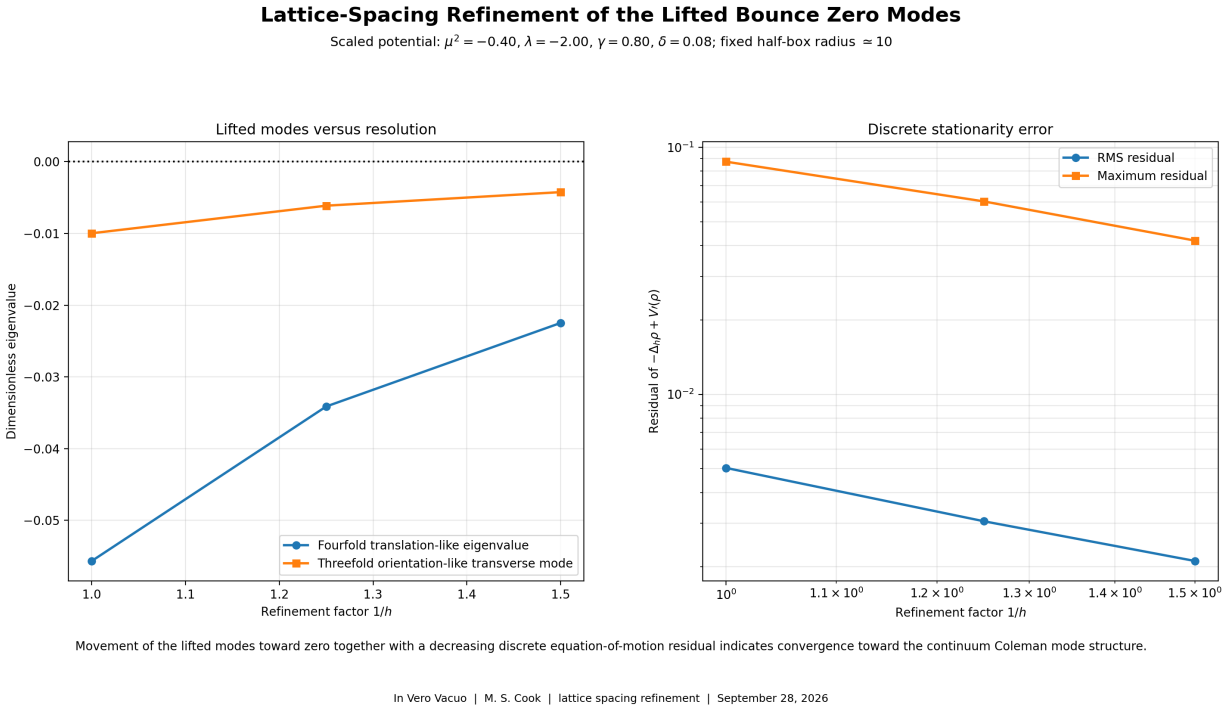


Figure 13.1: Spacing-refinement diagnostics for the sampled stationary bounce.

Lattice-Spacing Refinement Numerical Summary

Scaled potential: $\mu^2 = -0.40$, $\lambda = -2.00$, $\gamma = 0.80$, $\delta = 0.08$; fixed half-box radius $\simeq 10$

L	h	Boundary	Principal negative	Translation mean	Orientation	EOM RMS
20	1.000000	0.2820%	-2.357758e-01	-5.566158e-02	-9.976241e-03	5.032e-03
26	0.800000	0.2060%	-2.162747e-01	-3.412852e-02	-6.134877e-03	3.060e-03
32	0.666667	0.1673%	-2.058251e-01	-2.248334e-02	-4.244266e-03	2.114e-03

Eigenvalues are dimensionless in the reference coordinate system. The translation value is fourfold by hypercubic axis symmetry; the orientation value occurs in three identical internal components.

In Vero Vacuo | M. S. Cook | lattice spacing refinement | September 28, 2026

Figure 13.2: Evolution of the low collective sectors under lattice refinement. The study demonstrates why exact discrete stationarity is preferable to manually declaring shifted modes to be zero.

Chapter 14

End-to-End Numerical Result

14.1 Result ledger

Table 14.1: Principal numerical outputs of the production pipeline.

Stage	Quantity	Result
Continuum bounce	Center field	$\rho(0) = 1.367393286$
Continuum bounce	Boundary residual	0.281955%
Continuum bounce	Action	$B = 145.203947$
Discrete relaxation (regenerated)	RMS EOM residual	4.43266×10^{-12}
Discrete relaxation (regenerated)	Maximum EOM residual	6.30906×10^{-11}
Hessian	Physical negative mode	-0.1850415 in lattice units
Hessian	Translation sector	four modes at -0.00212653
Hessian	Orientation sector	three copies of a mode at -5.33×10^{-12}
SLQ	Bare trace–log	-2079.9042 ± 6.4504
Scalar continuum calculation	Reported trace–log	40.9324; 0.03 hardcoded estimate
Continuum matching	Determinant factor	1.2931×10^{-9}
Semiclassical expression	Translation factor included	5.9974×10^{-70} in lattice energy units^4
SU(2) prerequisites	Lattice Ward and homogeneous tests	PASS at $\xi = 0.5, 1, 2$
SU(2) $L = 26$ recovery	Bounce action and RMS residual	139.9422784742 and $4.4316323 \times 10^{-12}$
SU(2) stationary Ward test	Largest on-shell residual	5.52×10^{-14}; PASS
SU(2) regulated determinant	Exact homogeneous ξ scan	PASS as IR shift is removed
SU(2) counterterms	Scalar and full Feynman-gauge pole closure	PASS to 3.11×10^{-15}
October 5 $L = 6$ follow-up	4096-probe homogeneous trace	-20.75094 ± 1.62888; exact -19.56733
October 5 complete multiplets	Best tested constants, 5/1	12.51% variance reduction versus fresh 8/8
October 5 nonconstant audit	$L = 3, 4$ deterministic prerequisites	PASS at 10^{-10}; all profiles off shell

Stage	Quantity	Result
October 5 nonconstant SLQ	Largest 80-step probe quadrature error	1.05×10^{-11}; sampling dominates
SU(2) nonlinear vertices	Independent generator/table audit	PASS at tested routings; vertex APIs implemented
SU(2) matching interface	Full-gauged finite constants	Interface implemented; full-scope constants remain unset
October 7 scalar-only matching	Six radial potential coefficients	Computed at the stated benchmark; not full-$SU(2)$ values
October 8 full-gauged Step 1	Fixed routings, projectors, selected vertex tensors	Independent audit PASS; first four-point gauge-linked pilot integrated
October 8 four-point pilot	Gauge-linked radial 4-point insertion sector	Regulated result and checks recorded; QMC/extrapolation errors remain pilot-level
Full-gauged matching	Complete gauge-sector finite coefficients	Not computed; full-matching flag remains false
Gauged prefactor	Physical determinant/rate	Not yet calculated

14.2 Decay-rate status

Inserting the computed action and determinant factor into the scalar part of Eq. (1.2) gives

$$\left(\frac{B}{2\pi}\right)^2 \left|\frac{\det{}' H_B}{\det H_F}\right|^{-1/2} e^{-B} \simeq 5.9974 \times 10^{-70} \quad \text{(lattice energy units)}^4. \tag{14.1}$$

This includes the four translation collective coordinates through the standard factor $(B/2\pi)^2$ [2]. It does not include the remaining three-dimensional internal-orientation group-volume Jacobian. This expression remains a global-scalar partial result. It must not be relabelled as a gauged rate or multiplied by the new diagnostic gauge spectra. The gauge-orbit measure and false-vacuum gauge/ghost zero modes require their own consistent treatment.

Chapter 15

Output, Checkpointing, and Reproducibility

15.1 Checkpoint strategy

Long calculations are resumable at two levels:

- the relaxed field and Newton history are saved in a compressed NPZ file;
- in the established scalar SLQ mode, every completed stochastic probe is appended to a CSV checkpoint.

Low eigenpairs are cached separately because they are expensive and dominate memory. Random seeds are fixed through environment variables so that repeated runs can be compared probe by probe.

SU(2) determinant restart limitation

The present `su2_determinant` implementation writes its probe table only after all requested gauge parameters and probes have completed. Before a long cloud run, it should be extended to save low eigensystems and append each completed (ξ, probe) result atomically. This is a reliability requirement rather than a change in the determinant definition.

15.2 Cloud-migration checklist

For migration to a persistent compute environment, archive or recreate the following items before launching the production determinant:

1. the exact unified Python source and its SHA-256 digest;
2. the regenerated $L = 26$ NPZ checkpoint and a checksum stored outside the ephemeral workspace;
3. `su2_prefactor_validation_20261002/validation_summary.json` and its validation PDF;
4. the scalar relaxation and SLQ summaries used as regression references;
5. the counterterm coefficient template and all environment-variable settings; and
6. an output directory on persistent storage for low modes, per-probe checkpoints, summaries, and logs.

After transfer, rerun the inexpensive Ward suite and compare the bounce action, RMS residual, field checksum, and homogeneous gauge spread before beginning the production SLQ calculation.

15.3 Primary output files

Table 15.1: Machine-readable and publication outputs.

Path	Purpose
`discrete_bounce_relaxation_L26_h0.8/relaxed_bounce_checkpoint.npz`	Relaxed lattice field and Newton history
`discrete_bounce_relaxation_L26_h0.8/discrete_relaxation_summary.json`	Stationarity and low-spectrum summary
`relaxed_bounce_slq_L26_h0.8/probe_checkpoint.csv`	Resumable per-probe trace–log estimates
`relaxed_bounce_slq_L26_h0.8/slq_summary_by_steps.csv`	Lanczos-order convergence table
`relaxed_bounce_slq_L26_h0.8/relaxed_slq_summary.json`	Final bare SLQ result
`scaled_potential_continuum_renormalized/determinant_zero_mode_convergence.csv`	Regulated zero-mode extrapolation
`scaled_potential_continuum_renormalized/determinant_summary.json`	Continuum-renormalized determinant result
`Higgs_Continuum_Renormalized_Determinant_Report_L26.pdf`	Five-page concise production report
`su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz`	Target regenerated field; absent after the October 2 workspace reset and therefore requiring regeneration or external restoration
`su2_prefactor_validation_20261002/validation_summary.json`	October 2 PASS record for the regenerated stationary bounce
`su2_prefactor_validation_20261002/SU2_Prefactor_Validation_Report.pdf`	Gauge-identity and homogeneous-normalization plots
`su2_determinant_scan/determinant_summary.json`	Regulated gauge–Goldstone–ghost determinant status and gauge spreads
`su2_determinant_scan/determinant_summary.csv`	Determinant means, errors, IR shifts, and zero-mode policies
`su2_determinant_scan/SU2_Determinant_Gauge_Scan.pdf`	R_ξ and IR-regulator diagnostic plots
`su2_counterterm_matching/counterterm_basis.csv`	Local kinetic and polynomial basis integrals on the selected background
`su2_counterterm_matching/counterterm_summary.json`	Pole-closure tests and any supplied finite matching contribution
`su2_counterterm_coefficients_template.json`	Derived pole residues and explicit placeholders for finite matching constants
`SU2_Determinant_Counterterm_Morning_Report_20261001.tex`	Standalone October 1 determinant and counterterm development report
`SU2_Physical_Prefactor_Progress_Report.tex`	Standalone report wrapper, sharing the body reproduced in this guide
`SU2_Physical_Prefactor_Progress_Content.tex`	Complete progress report and adopted EFT matching prescription
`SU2_Physical_Prefactor_Progress_Report.pdf`	Compiled five-page standalone progress report

Path	Purpose
SU2_L6_Single_Core_Pilot_Narrative_Report_20261005.tex	October 5 pilot and recorded increased-probe/quadrature audit
SU2_L6_Complete_Multiplet_Report_20261005.tex	Recorded multiplet-coverage and matched-probe variance comparison
SU2_Nonconstant_Prerequisite_Report_20261005.tex	Fresh $L = 3, 4$ determinant, Ward, and local-basis audit report
su2_nonconstant_prerequisite_audit.py	Separate single-core audit harness; production solver unchanged
su2_nonconstant_prerequisites_20261005/audit_summary.json	Retained aggregate audit results; full raw archive only partially retained
SU2_Matching_Coefficient_Progress_Report_20261006.tex	October 6 audit of matching readiness and explicit finite-coefficient status
SU2_Next_Steps_Finite_Matching_20261006.tex	Corrected matching procedure and first scalar tadpole calculation
su2_lattice_vertices.py	Nonlinear scalar-link, Wilson-gauge, ghost, and Faddeev–Popov generators
su2_nonlinear_vertex_audit.py	Independent numerical audit of the nonlinear vertex implementation
su2_nonlinear_vertex_tables_20261007_conserving.json	Momentum routings, vertex tensor dimensions, and table metadata
su2_full_gauged_projection_setup.py	October 8 fixed routing, projector, and common-IR setup generator
su2_full_gauged_projection_tables_20261008.npz	Selected full color/polarization vertex tensors and projector arrays
su2_full_gauged_projection_tables_20261008.json	Fixed momenta, parameters, projector definitions, and IR prescription
su2_full_gauged_projection_audit.py	Independent vertex, kinematic, projector, and Ward-identity audit
su2_full_gauged_projection_audit_20261008.json	Retained PASS status, discrepancies, and tolerances
su2_four_point_gauge_sector_20261008.py	Gauge-linked four-Higgs insertion evaluator at the fixed local-contact routes
su2_four_point_gauge_sector_20261008_ircheck.json	Four-regulator pilot values, QMC errors, extrapolation check, and diagnostics

15.4 Recommended commands

```
# Fast analytic check
HIGGS_RUN_MODE=validation python Higgs_fluctuation_solver_stabilized.py

# Low-spectrum exploratory report
HIGGS_RUN_MODE=research HIGGS_L=20 HIGGS_NUM_MODES=132 \
python Higgs_fluctuation_solver_stabilized.py

# Individual production stages
HIGGS_RUN_MODE=relax python Higgs_fluctuation_solver_stabilized.py
HIGGS_RUN_MODE=slq python Higgs_fluctuation_solver_stabilized.py
HIGGS_RUN_MODE=continuum HIGGS_MU_SQ=-0.40 HIGGS_LAMBDA=-2.00 \
HIGGS_GAMMA=0.80 HIGGS_DELTA=0.08 HIGGS_L=20 \
```

```
python Higgs_fluctuation_solver_stabilized.py

# Complete scalar-only sequential production pipeline
HIGGS_RUN_MODE=full python Higgs_fluctuation_solver_stabilized.py

# New gauge prerequisites (not a physical prefactor)
HIGGS_RUN_MODE=su2_validation python Higgs_fluctuation_solver_stabilized.py

# Optional stationary-bounce gauge checks
HIGGS_RUN_MODE=su2_validation \
HIGGS_SU2_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_VALIDATION_OUTPUT=su2_prefactor_validation_20261002 \
python Higgs_fluctuation_solver_stabilized.py

# Regenerate the L=26 stationary field without duplicating its low spectrum
HIGGS_RUN_MODE=relax HIGGS_RELAX_L=26 HIGGS_RELAX_SPACING=0.8 \
HIGGS_RELAX_OUTPUT=su2_prefactor_bounce_L26_h0.8 \
HIGGS_RELAX_SKIP_SPECTRUM=1 \
python Higgs_fluctuation_solver_stabilized.py

# Regulated gauge--Goldstone--ghost determinant scan
HIGGS_RUN_MODE=su2_determinant \
HIGGS_SU2_DET_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_DET_OUTPUT=su2_determinant_scan_L26_h0.8 \
HIGGS_SU2_DET_XI=0.5,1,2 HIGGS_SU2_DET_METHOD=slq \
python Higgs_fluctuation_solver_stabilized.py

# Generate and independently audit nonlinear SU(2) vertex tables
HIGGS_RUN_MODE=su2_vertices python Higgs_fluctuation_solver_stabilized.py
python su2_nonlinear_vertex_audit.py

# Counterterm basis, pole closure, and supplied finite coefficients
HIGGS_RUN_MODE=su2_counterterms \
HIGGS_SU2_CT_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_CT_COEFFICIENTS=su2_counterterm_coefficients_template.json \
python Higgs_fluctuation_solver_stabilized.py

# Exploratory gauge spectrum; a Gaussian is diagnostic only
HIGGS_RUN_MODE=research HIGGS_SU2_GAUGE=1 HIGGS_SU2_G=0.65 \
HIGGS_SU2_XI=1 HIGGS_L=8 HIGGS_NUM_MODES=40 \
HIGGS_BACKGROUND_PROFILE=gaussian HIGGS_BOUNCE_WIDTH_SITES=1.2 \
HIGGS_OUTPUT_TAG=su2_diagnostic python Higgs_fluctuation_solver_stabilized.py
```

Listing 15.1: Typical commands for the unified solver.

Chapter 16

Limitations and Beyond-Standard-Model Extensions

16.1 What the present solver establishes

The completed calculation demonstrates a consistent chain from a metastable higher-order scalar potential to a stationary bounce, a matrix-free multi-component Hessian, collective-mode classification, a full bare lattice trace–log, and a finite continuum-renormalized determinant. This is already a substantial numerical test bed for scalar sectors beyond the renormalizable quartic Standard Model potential. The September 30 extension established a link-derived quadratic $SU(2)$ operator, exact homogeneous reference levels, and measured Ward-identity prerequisites on both small grids and a stationary bounce. The October 1 extension adds regulated coupled and ghost determinant backends, explicit false-vacuum constant-mode and R_ξ measure policies, a local EFT counterterm basis, and derived continuum pole residues through dimension 12. On October 2 the $L = 26$, $h = 0.8$ checkpoint was reproducibly regenerated and passed the stationary Ward tests, but the ephemeral workspace subsequently lost the untracked NPZ before the production determinant scan began. It still does not extend the scalar determinant values to a physical gauged rate because the finite lattice-to-$\overline{\mathrm{MS}}$ constants and controlled regulator limits remain incomplete.

The October 5 audits in Chapter 9 add genuine four-dimensional single-core off-shell tests: $L = 6$ homogeneous stochastic consistency, independent per-probe quadrature checks, complete-multiplet variance comparisons, and $L = 3, 4$ nonconstant dense-reference, Ward, and local pole-algebra checks. Complete constants reduced homogeneous variance modestly; larger low-mode windows were not automatically better. The fresh nonconstant tests passed their deterministic gates, but sampling uncertainty, finite matching, and physical-bounce prerequisites remain. They do not supersede the earlier scalar results with a gauged rate.

16.2 Comparison with Euclidean-time Monte Carlo approaches

Recent Euclidean lattice Monte Carlo studies have proposed alternative estimators for false-vacuum decay in quantum-mechanical systems [12, 13]. These methods share the Euclidean functional-integral setting of the present calculation, but their observables and numerical systematics differ from those of a semiclassical one-loop prefactor. Their existing demonstrations are in low-dimensional single-particle systems; extensions to quantum field theory are discussed in the cited work, but gauge-theory implementations have not yet been demonstrated.

Euclidean probability-density method. Shen, Draper, and El-Khadra define a decay-rate estimator from the probability density at the classical turning point b, schematically

$$\Gamma \simeq \sqrt{m}\, p(b, T), \tag{16.1}$$

where $p(b, T)$ is measured with constrained Euclidean path integrals [12]. In the window $1/m \ll T \ll 1/\Gamma$, the relevant probability density becomes approximately independent of T, making the required continuation approximately innocuous under the assumptions of the construction. A cut on a path functional $S_V[x]$ controls the unstable direction and suppresses configurations that evolve deeply into the true-vacuum region. The method does not require an explicit numerical evaluation of the bounce determinant, although semiclassical information is used in motivating and interpreting parts of the estimator.

Implicit decay amplitude and spectral reconstruction. Jin and Swaim instead formulate an implicit Fermi's-Golden-Rule-type expression,

$$\Gamma = 2\pi \langle FV | (H_{\mathrm{FV}} - H)\, \delta(H_{\mathrm{TV}} - E_{\mathrm{FV}})\, (H_{\mathrm{FV}} - H) | FV \rangle, \tag{16.2}$$

and infer the required matrix element from Euclidean correlators and spectral reconstruction [13]. A multi-ensemble sampling strategy addresses the exponential suppression of false-vacuum configurations. The tested estimator does not use a semiclassical bounce expansion, but it introduces the inverse problem and associated systematics of reconstructing a spectral density from Euclidean data. Its numerical demonstrations are also currently limited to one-dimensional quantum-mechanical systems.

Relation to the present SU(2) program. These Monte Carlo approaches are complementary to the program developed here. The present source constructs a link-derived gauge–Goldstone Hessian and ghost operator, verifies discrete Ward identities and homogeneous normalization tests, evaluates regulated finite-lattice determinant ratios, and constructs the counterterm basis and continuum pole residues. It does not yet contain the finite-matched and extrapolated coupled determinant or a completed physical gauged prefactor. Those remaining quantities require a consistent treatment of gauge fixing, Faddeev–Popov ghosts, collective coordinates, local counterterms, and regulator limits, none of which has yet been implemented in the cited quantum-mechanical Monte Carlo demonstrations.

The semiclassical framework is therefore the presently implemented route to the weak-coupling renormalized one-loop SU(2) prefactor in this project. Euclidean-time Monte Carlo estimators remain valuable candidates for future nonperturbative cross-checks rather than being excluded in principle from a future gauge-theory generalization.

16.3 What remains for a gauged Higgs theory

A physical gauged-$SU(2)$ or electroweak calculation requires additional operators and consistency conditions:

1. production evaluation of the implemented regulated coupled and ghost determinants on the relaxed bounce;
2. finite lattice-to-$\overline{\mathrm{MS}}$ matching of the retained local counterterms, including the derivative term;
3. confirmation of gauge-fixing cancellation after matching and regulator limits, not only in the exact homogeneous validation;
4. continuum extrapolation at fixed physical volume and an independent infinite-volume study with the massless false-vacuum gauge sector;

5. for an electroweak extension, hypercharge and fermion sectors;
6. normalization of any remaining collective coordinates;
7. scale and scheme dependence of the complete one-loop rate.

Scope and interpretation

Adding a gauge coupling to the potential is not sufficient. A gauge-invariant decay rate requires the entire coupled fluctuation problem. The current global-$O(4)$ result is therefore best viewed as a controlled scalar benchmark and a foundation for the more complete calculation.

16.4 Natural next calculations

- repeat the relaxed SLQ calculation at several h values while keeping a fixed physical box, then test counterterm-matched lattice extrapolation;
- calculate the internal $O(4)/O(3)$ collective-coordinate measure for the global model;
- factor the unified source into tested library functions while preserving the single-file production entry point;
- add automated regression tests comparing validation spectra, residuals, and determinant summaries;
- add atomic per-probe and per-ξ restart files to the `su2_determinant` mode;
- regenerate or externally restore the validated $L = 26$, $h = 0.8$ stationary checkpoint, preserve its checksum on persistent storage, and run a single-ξ pilot before the full deflated scan at $\xi = 0.5, 1, 2$;
- derive the finite lattice-to-$\overline{\mathrm{MS}}$ constants from homogeneous momentum integrals and a derivative expansion, then perform continuum and volume convergence checks.

Appendix A

Environment Variable Reference

Table A.1: Principal environment variables accepted by the unified program.

Variable	Typical value	Meaning
HIGGS_RUN_MODE	full	Select execution mode
HIGGS_SU2_GAUGE	1	Enable research-mode gauge diagnostics; rejected in scalar production modes
HIGGS_SU2_G	0.65	SU(2) gauge coupling for research diagnostics
HIGGS_SU2_XI	0.5, 1, 2	Positive background gauge parameter
HIGGS_SU2_NUM_MODES	12	Requested modes in each diagnostic generator/ghost window
HIGGS_SU2_VALIDATION_OUTPUT	directory	Output directory for SU(2) prerequisite tests
HIGGS_SU2_BOUNCE_FILE	NPZ path	Optional scaled L=26, h=0.8 stationary-bounce test input
HIGGS_SU2_DET_BOUNCE_FILE	NPZ path	Relaxed bounce used by the regulated gauge determinant
HIGGS_SU2_DET_OUTPUT	directory	Determinant CSV, JSON, low-mode, and PDF output directory
HIGGS_SU2_DET_METHOD	slq or exact	Select stochastic or small-matrix determinant backend
HIGGS_SU2_DET_XI	0.5,1,2	Gauge parameters included in the determinant scan
HIGGS_SU2_DET_IR_SHIFTS	0.001,0.003,0.01	Positive auxiliary mass-squared regulators
HIGGS_SU2_DET_STEPS	30,60	Lanczos orders for the determinant scan
HIGGS_SU2_DET_PROBES	8	Common Rademacher probes at each gauge parameter
HIGGS_SU2_DET_DEFLATED_MODES	12	Exact low modes removed before SLQ
HIGGS_SU2_CT_BOUNCE_FILE	NPZ path	Background used to evaluate local counterterm invariants
HIGGS_SU2_CT_OUTPUT	directory	Counterterm basis and matching output directory
HIGGS_SU2_CT_COEFFICIENTS	JSON path	Pole and optional finite matching coefficients
HIGGS_SU2_CT_SCALE	$\sqrt{0.4}$	$\overline{\mathrm{MS}}$ matching scale in solver units
HIGGS_SU2_VERTEX_SPACING	0.8	Lattice spacing for vertex tables
HIGGS_SU2_VERTEX_G0	0.65	Bare gauge coupling for vertex tables
HIGGS_SU2_VERTEX_OUTPUT	workspace-relative NPZ path	Vertex arrays; JSON sidecar uses the same basename
HIGGS_MU_SQ	-0.40	Quadratic coefficient convention
HIGGS_LAMBDA	-2.00	Quartic coupling
HIGGS_GAMMA	0.80	Sixth-order coupling
HIGGS_DELTA	0.08	Eighth-order coupling

Variable	Typical value	Meaning
HIGGS_L	20	Research/continuum reference size
HIGGS_NUM_MODES	120–132	Requested low spectral modes
HIGGS_BACKGROUND_PROFILE	stationary	Stationary or Gaussian profile
HIGGS_RELAX_L	26	Relaxation lattice length
HIGGS_RELAX_SPACING	0.8	Relaxation grid spacing
HIGGS_RELAX_MAX_NEWTON	14	Maximum Newton steps
HIGGS_RELAX_SKIP_SPECTRUM	1	Skip duplicate spectrum in full mode
HIGGS_SLQ_L	26	SLQ lattice length
HIGGS_SLQ_SPACING	0.8	SLQ lattice spacing
HIGGS_SLQ_PROBES	16	Rademacher probes
HIGGS_SLQ_DEFLATED_MODES	12	Exact low eigenpairs per channel
HIGGS_SLQ_STEPS	20,40,60	Lanczos orders
HIGGS_PARTIAL_WAVE_LMAX	70	Maximum continuum partial wave
HIGGS_RADIAL_CUTOFF	20	Continuum radial cutoff
HIGGS_DETERMINANT_OUTPUT	directory	Continuum output directory

Appendix B

Section Map of the Unified Source

Table B.1: Guide from source-code regions to numerical responsibilities. Line numbers refer to the October 8, 2026 workspace versions.

Approximate lines/file	Section	Responsibility
1–227	Imports and SU(2) shared operators	Imports nonlinear vertex APIs; modal export and quadratic link-Hessian tools
228–329	SU(2) reference operators	Homogeneous spectra and exact momentum multiplicities
330–582	SU(2) validation	Ward identities, decoupling, normalization, and optional bounce checks
583–1091	SU(2) determinant	Exact/SLQ trace–logs, zero-mode policies, and gauge scan
1092–1427	SU(2) counterterms	Local basis, pole residues, coefficient interface, and report
1428–1561	Vertex-table mode	Configured nonlinear SU(2) momentum/color table generation
1562–1619	Dispatcher/guard	Ten-mode dispatch and sequential scalar pipeline
1620–2800	Scalar production stages	Relaxation, SLQ, and continuum calculation modes
2801–2982	Physics/configuration	Units, scalar potential, and run settings
2983–3187	Operators and validation	Scalar Hessians, SU(2) wrappers, Fourier spectrum, exact checks
3188–3462	Background	Stationary/Gaussian profiles and bounce setup
3463–3754	Hessian and spectral extraction	Operator assembly, low modes, and regularized spectral ratio
3755–4204	Diagnostics/reporting	Spectrum visualization and scalar/gauge report outputs
4205–end	Console summary	Scalar results and optional gauge diagnostic status
`su2_lattice_vertices.py`	Nonlinear vertex library	Scalar-link, Wilson-gauge, gauge-ghost, scalar-ghost, and FP-matrix APIs
`su2_nonlinear_vertex_audit.py`	Independent audit	Separate action expansions, finite transformations, and vertex-table comparisons

Appendix C

Compact Algorithm Summary

Step 1: Read the selected execution mode and numerical parameters.

Step 2: Validate the periodic matrix-free operator on an exactly solvable test.

Step 3: Locate the false vacuum, barrier, and deeper nonzero minimum.

Step 4: Shoot the continuum $O(4)$ bounce using overshoot/undershoot bisection.

Step 5: Sample the bounce on a periodic L^4 lattice.

Step 6: Relax the sample until the discrete equation of motion is satisfied.

Step 7: Construct longitudinal, transverse, and false-vacuum Hessian operators.

Step 8: Compute and classify the low negative and collective-coordinate modes.

Step 9: Deflate these modes and estimate the full bare trace–log with SLQ.

Step 10: Evaluate the continuum partial-wave determinant with asymptotic subtraction and a regulated primed zero-mode limit.

Step 11: Combine the action, translation factor, and renormalized determinant, while clearly retaining the missing internal-orientation normalization.

Step 12: Write checkpoints, data tables, figures, JSON summaries, and PDF reports.

The steps above describe the scalar pipeline. The separate nonlinear-vertex mode generates and audits link, plaquette, and ghost rules; it does not feed the scalar pipeline or gauged determinant. The quadratic gauge branch builds the link Hessian and ghost operator, runs Ward, Fourier, and normalization checks, and can report exploratory spectra. The October 1 modes further evaluate regulated determinant ratios, separate false-vacuum constant-mode policies, construct the local counterterm basis, and verify continuum pole closure. Finite lattice-to-$\overline{\text{MS}}$ matching and controlled continuum and volume limits remain future work under the conditions in the following appendix. The October 5 test evidence and its provenance are incorporated in Chapter 9; the scalar-only radial coefficients are now available, but the full-gauged matching constants remain unset despite the successful vertex prerequisites.

Appendix D

SU(2) Physical Prefactor Progress Report

The complete foundational report generated on September 30, 2026 is reproduced below as native LaTeX, including its EFT matching prescription. The determinant and counterterm implementation added on October 1 is documented in the new chapter above and in `SU2_Determinant_Counterterm_Morning_Report_20261001.tex`. The September 30 source body is shared with `SU2_Physical_Prefactor_Progress_Report.tex`; no external compiled PDF is required to build this appendix.

Status: prerequisites verified; physical gauged prefactor not yet calculated. This report records newly executed lattice tests and bounce relaxation. It does not convert a diagnostic spectrum or a bare determinant into a physical rate. No gauged renormalized determinant is reported.

D.1 Confirmed benchmark and reproducibility

The target theory is a pure $SU(2)$ gauge theory coupled to one Higgs doublet, with vanishing background gauge field and radial amplitude ρ. Hypercharge and fermions are excluded. The benchmark is

$$V(\rho) = 0.2\rho^2 - 0.5\rho^4 + \frac{0.8}{6}\rho^6 + 0.01\rho^8, \qquad g = 0.65, \qquad \mu_R = \sqrt{0.4}.$$

These are the scaled production units, not an independently specified physical GeV normalization. A rate quoted in GeV^4 requires an explicit conversion scale. The exploratory research spacing $a = 1/500\,\mathrm{GeV}^{-1}$ must not silently be identified with the production spacing $h = 0.8$.

Quantity	New calculation	Meaning
Lattice	26^4, $h = 0.8$	Periodic scaled box
Scalar sites	456,976	One component
Bosonic dimension	7,311,616	Radial plus three five-field blocks
Discrete action	139.9422784742	Not the continuum action
EOM RMS residual	4.43266×10^{-12}	Regenerated stationary field
EOM maximum residual	6.30906×10^{-11}	Regenerated stationary field
Gauge parameters tested	$0.5, 1, 2$	Background R_ξ gauge

Table D.1: Executed stationary-bounce prerequisites.

The regenerated field is saved at

`su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz`

The reproducible tests are embedded in `Higgs_fluctuation_solver_stabilized.py`, under the new `su2_validation` mode. The machine-readable test record is `su2_prefactor_validation/validation_summary.json`.

D.2 Lattice Hessian derived from links

The earlier site-centered continuum stencil has been replaced by the quadratic expansion of the covariant nearest-neighbor scalar action and the zero-background gauge-link action. This change matters because self-adjointness alone does not enforce a lattice gauge identity. Gauge components $A_{x\mu}$ occupy outgoing links; χ_x occupies sites. Define forward differences G_μ and backward divergence D by

$$(G_\mu f)_x = \frac{f_{x+\hat{\mu}} - f_x}{h}, \quad (DA)_x = \sum_\mu \frac{A_{x\mu} - A_{x-\hat{\mu},\mu}}{h}, \quad G^\dagger = -D, \quad K = -DG.$$

For each generator, with $q = g/2$, the gauge-fixing functional is $F = DA - \xi q \rho \chi$. The quadratic scalar-link contribution is

$$S^{(2)}_{\text{link}} = \frac{h^4}{2} \sum_{x,\mu} \left[\frac{(\chi_{x+\hat{\mu}} - \chi_x)^2}{h^2} + q^2 \rho_x \rho_{x+\hat{\mu}} A^2_{x\mu} \right] \tag{D.1}$$

$$+ h^4 \sum_{x,\mu} \frac{q}{h} A_{x\mu} (\rho_{x+\hat{\mu}} \chi_x - \rho_x \chi_{x+\hat{\mu}}). \tag{D.2}$$

Adding the quadratic gauge action, scalar potential curvature, and $h^4 \sum_x F_x^2/(2\xi)$ gives the implemented block

$$H_{A\chi} = \begin{pmatrix} K\mathbf{1} + (1-\xi^{-1})GD + M^2_{\text{link}} & C \\ C^\dagger & K + U_T + \xi q^2 \rho^2 \end{pmatrix}, \qquad H_c = K + \xi q^2 \rho^2,$$

where $U_T = V\prime(\rho)/\rho$ is understood by its smooth polynomial extension at $\rho = 0$,

$$(M^2_{\text{link}})_{x\mu} = q^2 \rho_x \rho_{x+\hat{\mu}}, \quad (C\chi)_{x\mu} = \frac{q}{h}(\rho_{x+\hat{\mu}} - \rho_x)(\chi_x + \chi_{x+\hat{\mu}}).$$

The transpose $C^\dagger$ is implemented explicitly. In the continuum limit $C\chi \to 2q(\partial_\mu \rho)\chi$, recovering the gradient mixing of the gauge–Goldstone system [11]. The radial block remains $K + V\prime\prime(\rho)$.

D.3 Gauge identities and measured checks

For a site gauge parameter α, the gauge direction is $R\alpha = (G\alpha, q\rho\alpha)$. The discrete off-shell identity is

$$H_{A\chi} R\alpha = \begin{pmatrix} \xi^{-1} G(H_c \alpha) \\ q\rho H_c \alpha \end{pmatrix} + \begin{pmatrix} 0 \\ q\alpha \left[K\rho + V\prime(\rho) \right] \end{pmatrix}.$$

The residual term must be retained on an arbitrary background. It vanishes on a stationary bounce; it must not be discarded on the old Gaussian diagnostic profile. The implemented tests check both forms.

The $g \to 0$, $\xi = 1$ decoupling test gives zero measured relative error. Homogeneous coupled and ghost spectra agree with complete Fourier levels to relative errors below 9.3×10^{-16}. False-vacuum research spectra now use these exact levels, preserving momentum degeneracies that the old single-vector ARPACK calculations did not always recover.

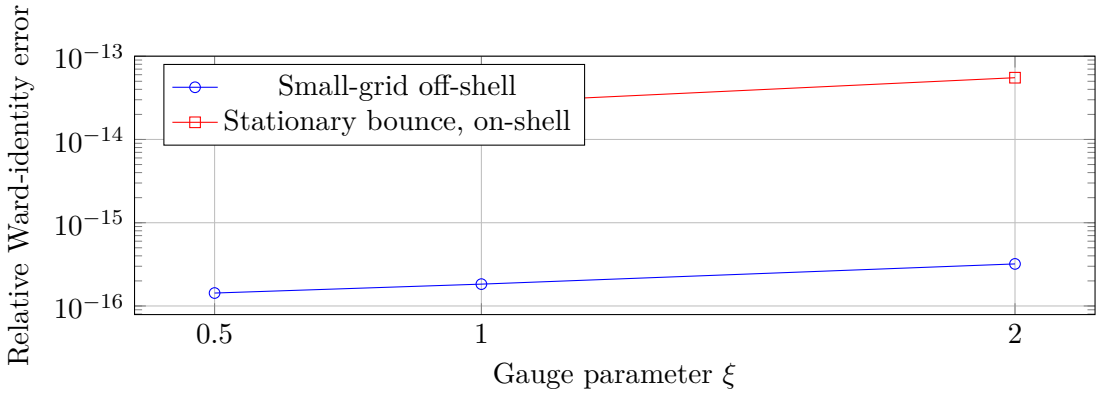
For a constant stationary broken background, the normalized one-generator one-loop expression

$$\frac{1}{2} \log\det H_{A\chi} - \log\det H_c + \frac{N}{2} \log \xi$$

ξ	Small-grid off-shell error	Bounce off-shell error	Bounce on-shell error
0.5	1.43×10^{-16}	1.41×10^{-16}	1.38×10^{-14}
1.0	1.83×10^{-16}	1.62×10^{-16}	2.76×10^{-14}
2.0	3.20×10^{-16}	2.78×10^{-16}	5.52×10^{-14}

Table D.2: Relative vector-norm errors. The on-shell error reflects the nonzero, tightly converged lattice EOM residual.

is 296.100090843863 at all three tested gauges, with a spread of 5.12×10^{-13}. This tests the gauge-fixing normalization in a positive, finite-volume example. It is not a false-vacuum/bounce ratio and is not a gauged-prefactor gauge-independence test.



D.4 New renormalization issue: the potential is not closed

The previously proposed prescription specified the sixth- and eighth-order Wilson coefficients, but it did not specify all local operators generated when those terms are included nonperturbatively in a one-loop determinant. Already the scalar part of the local divergence is proportional to $U_L^2 + 3U_T^2$, where

$$U_L = m_F^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \quad U_T = m_F^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6.$$

Direct symbolic expansion gives

$$U_L^2 + 3U_T^2 = 4m_F^4 + 12\lambda m_F^2\rho^2 + (12\lambda^2 + 16\gamma m_F^2)\rho^4 \tag{D.3}$$

$$+ (36\lambda\gamma + 20\delta m_F^2)\rho^6 + (28\gamma^2 + 48\lambda\delta)\rho^8 \tag{D.4}$$

$$+ 76\gamma\delta\rho^{10} + 52\delta^2\rho^{12}. \tag{D.5}$$

The last two terms require higher potential counterterms under this nonperturbative treatment. Their finite renormalized coefficients are not fixed just by declaring γ and δ to be $\overline{\text{MS}}$ coefficients. Gauge fluctuations additionally require the appropriate derivative/field-normalization counterterm [11]. This algebra is a counterterm-closure diagnostic, not a complete $SU(2)$ matching computation.

D.4.1 EFT matching prescription and counterterm closure

The scalar contribution to the one-loop local divergences contains $U_L^2 + 3U_T^2$. Its expansion above generates ρ^{10} and ρ^{12} terms absent from the original production potential. This establishes the need for additional local potential counterterms when the sixth- and eighth-order interactions are retained nonperturbatively in the leading fluctuation operators. It does not identify the complete gauge–Goldstone–ghost divergence: the gauged calculation also requires its potential and derivative contributions [11].

Gauge-invariant definition and radial restriction. Let $\mathcal{H}$ denote the Higgs doublet and define $\rho = \sqrt{2\mathcal{H}^\dagger\mathcal{H}}$. The invariant Euclidean Lagrangian relevant to the benchmark is

$$\mathcal{L}_{\rm inv} = Z_{\mathcal{H}}(\mu_R)(D_\nu\mathcal{H})^\dagger D_\nu\mathcal{H} + \frac{1}{4}F^a_{\nu\sigma}F^a_{\nu\sigma} + V_{\rm EFT}(\rho;\mu_R), \tag{D.6}$$

where D_ν is the $SU(2)$ covariant derivative. Gauge-fixing and Faddeev–Popov ghost terms are added to the quantized action. On the radial background with fixed internal orientation and vanishing background gauge field, the scalar part reduces to

$$\mathcal{L}_{\rm radial} = \frac{1}{2}Z_{\mathcal{H}}(\mu_R)(\partial_\nu\rho)^2 + V_{\rm EFT}(\rho;\mu_R).$$

The radial expression alone is therefore a background restriction, not the full gauged EFT.

Use $m^2 = -\mu^2$ to distinguish the positive false-vacuum mass coefficient from the source-code convention. The renormalized potential is written as

$$\begin{aligned} V_{\rm EFT}(\rho;\mu_R) = {}& \frac{1}{2}m^2(\mu_R)\rho^2 + \frac{1}{4}\lambda(\mu_R)\rho^4 + \frac{1}{6}\gamma(\mu_R)\rho^6 + \frac{1}{8}\delta(\mu_R)\rho^8 \\ & + \frac{1}{10}c_{10}(\mu_R)\rho^{10} + \frac{1}{12}c_{12}(\mu_R)\rho^{12}. \end{aligned} \tag{D.7}$$

1. Closure at the specified loop order. For this benchmark c_{10} and c_{12} are counted as *one-loop-order* coefficients. They enter the local one-loop action, but are not inserted into the leading bounce equation or leading Hessian used inside the one-loop determinant. Inserting them there would contribute beyond the retained loop order. If they were instead treated as independent nonperturbative leading interactions, the enlarged potential would not be closed at dimension 12: for example, the curvature of a ρ^{12} interaction is proportional to ρ^{10}, whose square generates ρ^{20}.

At the retained order, the local potential counterterms must include those for $m^2, \lambda, \gamma, \delta, c_{10}, c_{12}$, together with the covariant kinetic counterterm $\delta Z_{\mathcal{H}}$ needed for the bounce-background effective action. The field-independent vacuum term cancels in the bounce-minus-false-vacuum action. This is a closure prescription for the specified background sector and loop order, not a claim that these terms alone renormalize every amplitude of the full gauge theory. Gauge-field and gauge-coupling renormalization, and any further operators required by the chosen EFT expansion, must be treated consistently where they contribute.

2. Finite matching at the reference scale. We adopt $M_R = \sqrt{0.4}$ in the scaled production units and impose the renormalized benchmark conditions

$$Z_{\mathcal{H}}(M_R) = 1, \qquad c_{10}(M_R) = c_{12}(M_R) = 0, \tag{D.8}$$

$$m^2(M_R) = 0.4, \qquad \lambda(M_R) = -2.00, \qquad \gamma(M_R) = 0.80, \qquad \delta(M_R) = 0.08, \qquad g(M_R) = 0.65. \tag{D.9}$$

The required poles are to be subtracted in the $\overline{\rm MS}$ prescription. Canonical normalization here defines the renormalized local kinetic coefficient at the reference scale; it does not assert that the divergent bare counterterm vanishes. The vanishing higher-operator coefficients fix additional finite matching conditions rather than following from the original four potential parameters.

These numerical conditions reproduce the *tree-level* production potential, its barrier, and its vacuum depths. They do not impose that the loop-corrected barrier height, false-vacuum curvature, and true-vacuum depth remain unchanged. Such requirements would define additional finite renormalization conditions and would need to be distinguished from the benchmark chosen here.

Wilson-coefficient dimensions and scale dependence. In four dimensions, for a dimensionful canonically normalized scalar, one may parameterize

$$c_{10}(\mu_R) = \frac{C_{10}(\mu_R)}{\Lambda^6}, \qquad c_{12}(\mu_R) = \frac{C_{12}(\mu_R)}{\Lambda^8},$$

with dimensionless C_{10}, C_{12}. These dimensional relations do not specify a UV completion or establish a numerical suppression in this scaled benchmark. No value of Λ or physical energy conversion has yet been supplied.

Running away from M_R is part of the retained-order renormalization group description, not itself an EFT truncation error. The running coefficients must compensate the explicit scale dependence of the loop effective action to the calculated order. A controlled estimate of omitted operators additionally requires power counting, the dimensionless Wilson coefficients, and checks of the bounce field amplitudes and gradient scales against Λ; a suppression by $(\rho/\Lambda)^2$ is not assumed without those checks.

This prescription defines a reproducible one-loop benchmark for the specified bounce-background sector. It fixes the additional finite matching conditions but does not, by itself, establish the EFT validity range or complete the renormalized gauged prefactor calculation. These conditions are adopted here as the calculation definition; their counterterm and determinant implementation remains part of the numerical work described below.

D.5 Remaining steps and safeguards

The complete one-loop gauge contribution uses the radial block, three coupled generator blocks, and three complex ghost determinants:

$$\Delta\Gamma_1 = \frac{1}{2}\log\left|\frac{\det{}' H_{r,B}}{\det H_{r,F}}\right| + \frac{3}{2}\log\left|\frac{\det H_{A\chi,B}}{\det H_{A\chi,F}}\right| - 3\log\left|\frac{\det H_{c,B}}{\det H_{c,F}}\right| + \Delta S_{\rm ct}.$$

This equation is schematic until gauge-volume and zero-mode conventions are fixed. In particular the symmetric false vacuum has constant gauge and ghost zero modes on the periodic box. Gauge-orbit normalization must be handled consistently; the old global-$O(4)$ orientation Jacobian cannot simply be multiplied into the gauged result. A consistent gauge-invariant treatment of the decay determinant and its renormalization is essential [10].

With the matching conditions defined above, work remains on coupled determinants, the false-vacuum zero modes, ultraviolet matching, the $h \to 0$ and volume limits, translation normalization, and uncertainty estimation. No Gaussian spectral window supplies these steps. The program now rejects `HIGGS_SU2_GAUGE=1` in scalar-only `full`, `slq`, `continuum`, and `relax` modes, so a scalar pipeline cannot silently masquerade as a gauge calculation. The old scalar continuum uncertainty 0.03 is hardcoded in the source; it is not a measured gauged error and is not reused here.

D.6 Commands for the completed prerequisites

```
HIGGS_RUN_MODE=relax HIGGS_RELAX_SKIP_SPECTRUM=1 \
HIGGS_RELAX_OUTPUT=su2_prefactor_bounce_L26_h0.8 \
python Higgs_fluctuation_solver_stabilized.py

HIGGS_RUN_MODE=su2_validation \
HIGGS_SU2_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
python Higgs_fluctuation_solver_stabilized.py
```

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